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Stellar spectral scans with the ALMA Compact Array

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ABSTRACT

Something will be added here

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1. INTRODUCTION

The circumstellar envelopes (CSEs) around stars on the Asymptotic Giant Branch (AGB) are known to host rich molecular environments. This is a consequence of the instabilities present in their stellar atmospheres leading to gas being ejected from them and then cooling down a few stellar radii away. Depending on the initial mass of the central star, AGB stars can develop CSEs with more oxygen than carbon (M stars, with all carbon locked in CO), more carbon than oxygen (C stars, with oxygen locked in CO) or comparable amounts of these elements (S stars). Coherently with such a picture, we expect to detect such molecules as SiO and H_2O in O-rich CSEs, HCN and CS in C-rich environments, and a mixture of these in S stars, while CO will trace all CSEs, regardless of their chemical classification.

This description can be complicated by a clumpy CSE morphology, which allows interstellar UV photons to penetrate the CSE, and by high-energy photons from the central star evolving toward the white-dwarf stage; both these circumstances can in fact alter the chemical inventory through the destruction of molecules, which frees up atoms of carbon (in O-rich CSEs) or oxygen (in C-rich CSEs), making them available for reactions. Also, shocks in the CSE linked to the propagation of high-velocity molecular outflows in bipolar nebulae can be detected. These also contribute to complicating the chemistry, because they can lead to removing chemical species from dust grains or breaking apart existing molecules, so that new ones can form. AGB and post-AGB CSEs are therefore active chemical laboratories that have deserved the attention of many studies over the years, in the attempt to figure out the chemical evolution of stars on and after the AGB.

The best way to achieve this goal is to carry out blind spectral-line surveys of as many such stars as possible. These works are typically time consuming and have targeted so far only a few bright sources per evolutionary stage, such as IRC+10 216, IK Tau, CRL 2688, and CRL 618.

In this paper, we report about a blind survey that we performed in 2019 with the Compact Array (ACA) of the Atacama Large Millimeter/sub-millimeter Array (ALMA), targeting a sample of stars classified as evolved objects or possible such objects in the range of right ascension between 19 and 08 hours. This range of right ascension was due

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Table 1. Source list

Source	Alternative Identifier(s)	α_{2000}	δ_{2000}	Parallax[a] (mas)	Radial Velocity[b] (km s^{-1})
2MASS J00235767-7205296	Cl∗ NGC 104 LW 8	00:23:57.57	-72:05:29.43	0.2398 [0.0326]	···
2MASS J01302276-7303339	GSC 09142-00333	01:30:22.77	-73:03:33.95	0.5095 [0.0114]	31.53 [0.22]
UU For	RAFGL 357	02:37:23.06	-26:58:42.32	1.6293 [0.1312]	6.0 [···]
R Hor	HD 18242	02:53:52.77	-49:53:22.73	4.3236 [0.1524]	60.00 [4.4]
V743 Mon	HD 50138	06:51:33.39	-06:57:59.45	2.8487 [0.0459]	34.0 [2]
IRAS 06531-0216	V918 Mon	06:55:40.26	-02:20:15.65	0.8633 [0.0977]	57.18 [2.61]
FS CMa	HD 45677	06:28:17.42	-13:03:11.12	1.7481 [0.0448]	21.6 [2]
IRAS 07145-1428	2MASS J07165246-1433587	07:16:52.46	-14:33:58.72	0.1919 [0.0446]	···
IRAS 07180-1314	RAFGL 5229	07:20:19.54	-13:20:13.88	0.856 [0.1911]	···
IRAS 08500-3254	RAFGL 5253	08:52:05.60	-33:05:44.02	0.655 [0.0735]	···
V510 Pup	IRAS 08005-2356	08:02:40.70	-24:04:42.58	2.2922 [0.3633]	···
IRAS 19336-0400	SS 441	19:36:17.51	-03:53:25.09	0.1063 [0.0162]	···
IRAS 19590-1249	BD-13 5550	20:01:49.82	-12:41:17.76	0.1517 [0.0261]	···
RZ Sgr	HD 192194	20:15:28.40	-44:24:37.47	2.3596 [0.0553]	···
RV Aqr	RAFGL 2702	21:05:51.73	-00:12:42.12	1.5742 [0.0582]	-1.0 [5]
DS Aqr	HD 216457	22:53:17.03	-18:35:30.97	0.1677 [0.0156]	-21.74 [2.18]
HM Aqr	HD 213985	22:35:27.52	-17:15:26.89	1.3853 [0.0558]	···

[a] From Gaia Collaboration (2020). Uncertainties are in square brackets. Gaia parallaxes may not be reliable representations of distance for AGB stars, since they can be on the order of the stellar diameter (e.g., Xu et al. 2019).

[b] From SIMBAD, (CHECK whether optical definition). Uncertainties are in square brackets.

to the nature of the project, which was proposed to fill up a scheduling gap in the ACA around 22–05 hours and was approved as an Observatory Project, without any proprietary time.

The targets selected for the study are listed in Table 1 and their characteristics are given in Table 2.

2. OBSERVATIONS AND RESULTS

Observations were performed within project 2018.A.00047.S with the ALMA Compact Array between 2019-06-20 and 2019-08-29 in bands 3 and 6, spanning continuously the frequency ranges ∼85–116 GHz and ∼211–272 GHz. The option to set up spectral scans was not offered in the ACA at the time of our project, therefore each observation was set up to sample a total of ∼7.5 GHz with two pairs of spectral windows containing two spectral windows each. In each pair, the two spectral windows overlapped by some 20 MHz and the inner spectral edges of the two pairs were about 8 GHz apart. Five spectral setups allowed us to span the entire frequency range in Band 3, while nine were necessary in Band 6.

The typical synthetic beam size ranged from ∼14.5″ to ∼11″ in band 3 and from ∼5.5″ to ∼4.5″ in band 6, with rms noise of ∼26 mJy beam^{-1} over ∼1.13 MHz in both bands, equivalent to ∼0.35 mJy beam^{-1} in the ∼7.5 GHz continuum images. We observed a total of 17 sources in band 3 and five of these were also observed in band 6.

The data were calibrated and imaged in CASA 5.4.0-70 with the ALMA pipeline version Pipeline-CASA54-P1-B. The data cubes and continuum images produced by the pipeline were retrieved from the ALMA archive and used without further processing.

Table 2. Source characteristics

Source	Type[a]		Variability[b]		Magnitude[a]
	Stellar	Spectral	Type	Period (days)	Range
2MASS J00235767-7205296	Post-AGB	···	···	···	···
2MASS J01302276-7303339	Post-AGB	···	···	···	···
UU For	Mira Variable	M7-M9	M	469.0	8.3 – <15.0 V
R Hor	Mira Variable	M5e-M8eII-III	M	407.6	4.7 – 14.3 V
V743 Mon	B[e] Star	B6/7III/V	FSCMa	···	6.53 - 6.87 V
IRAS 06531-0216	C star	C	M	595:	18.743 (3.111) g
FS CMa	B[e] star	B2IVep	FSCMa	···	7.35 - 8.58 V
IRAS 07145-1428	C star	···	VAR	···	14.86 (2.00) r
IRAS 07180-1314	OH/IR star	···	VAR	···	14.47 (4.74) i
IRAS 08500-3254	Mira Variable	Me	M	256.0	11.74 (3.4) V
V510 Pup	Post-AGB	F5Ie	SRD	···	11.7 – 12.2 V
IRAS 19336-0400	Post-AGB	BIe	PPN	···	12.5 (0.30) V
IRAS 19590-1249	Post-AGB	B1Ibe	$*$	···	11.08 – 11.51 V
RZ Sgr	S star	S4,4ep	M	207.8	6.95 – 10.61 V
RV Aqr	C star	C6,3e	M	453.0	-9.0 – <13 V
DS Aqr	RV Tauri Variable	F2-K1IIw	RVA	77.5	10.12 – 11.36 V
HM Aqr	Binary Post-AGB	A0III	E	259.6	8.72 – 9.40 V

[a]From the AAVSO and SIMBAD.

[b]Variability designations are those of the AAVSO. M = Mira Variable; FSCMa = B[e]-type stars that display irregular, long-term variations; VAR = Suspected variable of unspecified type; SRD = Semi-Regular Variable of F, G or K spectral type; PPN = Post-AGB stars embedded in pre-planetary nebulae and displaying semi-regular variability; $*$ = Variable stars outside of the usual classification ranges; RVA = RV Tauri Variable; E = Eclipsing Binary. In the case of HM Aqr (HD 213985), the variability period corresponds to the eclipsing binary orbital period of 259.6±0.7 days (Oomen et al. 2018).

In our small survey, we found continuum emission beyond 5σ in at least one frequency setup in ten of our targets, as summarized in Tables 3 and 4. Spectral lines were instead detected in nine targets.

FS CMa is the prototype of a group of variable stars and V743 Mon is another member of this group. These were sometimes considered as possible post-AGB objects, but the latest studies point to these stars being the product of stellar mergers, often surrounded by gas and dust. Both stars have been detected in continuum, but no spectral line was detected within the sensitivity limit of the present project. They display a spectral index of 1, confirmed for FS CMa by previous detections with the VLA and the JCMT (Rodríguez et al. 2012). The steep spectral index in V743 Mon is confirmed by the non-detection of this target at 3 GHz in the VLASS Survey. As pointed out by Rodríguez et al. (2012), such indices are indicative of a stellar wind with a radial density distribution of the gas steeper than the canonical r^{-2}. There is no indication of the presence of dust emission in the mm range, which may be explained by the amount of dust being so small that the wind emission dominates or by the fact that the dust is made by small hot grains, not emitting efficiently at mm wavelengths. If we look at the SEDs of these stars, which clearly show a wide infrared bump between 10 and 100 μm, the latter hypothesis seems plausible. Goran Sandell et al (2018, ApJ, 864, 104) modelled the SED of V743 Mon with a greybody, interpreting the result at 1.3 mm by Lee et al (2016) as emission from dust. They pointed out that the derived dust emissivity was especially low (0.12). Our data and the flux density value at 225 GHz found by Lee et al (2016) match very well, giving a spectral index very close to that of

Table 3. Continuum flux densities in Band 3 (3 mm) with their errors. For detections, the error equals $\sqrt{rms^2 + (0.05 \times S_\nu)^2}$, while for non-detections it equals the rms.

Source	S_ν (mJy)				
	92.94 GHz	96.66 GHz	100.39 GHz	104.12 GHz	107.85 GHz
R Hor	4.6 ± 0.7	3.6 ± 0.7	4.7 ± 0.6	5.3 ± 0.5	5.4 ± 0.4
V743 Mon	8.6 ± 0.5	8.6 ± 0.4	9.2 ± 0.4	10.2 ± 0.4	9.3 ± 0.7
FS CMa	22.1 ± 0.5	25.0 ± 0.5	24.7 ± 0.5	24.4 ± 0.6	26.3 ± 0.5
V510 Pup	0.7	0.4	0.3	0.5	2.0 ± 0.4
IRAS 19336-0400	7.9 ± 0.3	8.2 ± 0.3	7.9 ± 0.2	7.5 ± 0.2	7.5 ± 0.2
IRAS 19590-1249	2.8 ± 0.3	2.2 ± 0.3	2.2 ± 0.3	2.5 ± 0.3	2.6 ± 0.3
RV Aqr	0.5	0.3	0.4	1.8 ± 0.3	0.3

Table 4. Continuum detections in Band 6 (1.3 mm). For detections, the error equals $\sqrt{rms^2 + (0.05 \times S_\nu)^2}$, while for non detections it equals the rms.

Source	S_ν (mJy)								
	220.36 GHz	224.09 GHz	227.82 GHz	231.54 GHz	249.76 GHz	253.49 GHz	257.21 GHz	260.96 GHz	264.67 GHz
IRAS 06531-0216	2.4 ± 0.4	2.4 ± 0.3	2.6 ± 0.3	0.5	2.5 ± 0.3	2.7 ± 0.6	2.6 ± 0.4	2.6 ± 0.6	2.4 ± 0.5
IRAS 07180-1314	0.4	0.6	0.5	0.3	2.1 ± 0.3	0.5	0.6	0.5	2.2 ± 0.4
IRAS 08500-3254	0.5	0.3	0.5	0.3	0.4	0.8	0.4	2.4 ± 0.4	0.3
V510 Pup	7.3 ± 0.7	6.5 ± 0.7	6.4 ± 0.5	7.0 ± 0.4	9.0 ± 0.4	7.7 ± 0.5	9.2 ± 0.5	9.3 ± 0.6	10.1 ± 0.7

FS CMa. We therefore conclude that the emission at 3 and 1.5 mm in V743 Mon does not arise from dust, but is in fact due to stellar wind.

2.1. *Non-detections*

3. DISCUSSION

3.1. *Continuum SEDs*

The millimeter continuum of AGB stars may have both stellar and thermal dust contributions. The radiation from these two components follows two different spectral distributions, with the stellar one being a Planck function and the dust being approximated with a modified blackbody proportional to $\nu^{2+\beta}$ in the mm range, where β is the grain emissivity. In evolved stars, grain emissivity is typically found to be around 1 (**REFERENCE**).

In Figures 2 and 1, we display the SEDs of our targets with the data from our observations and catalog data from IRAS, AKARI, and Herschel, when available. For IRAS 19336-0400 and 19590-1249, we display low-frequency data from Cerrigone et al. (2011), for FS CMa from Rodríguez et al. (2012) and Di Francesco et al. (2008), and for V743 Mon from Lee et al. (2016).

ADD TWO MORE SOURCES

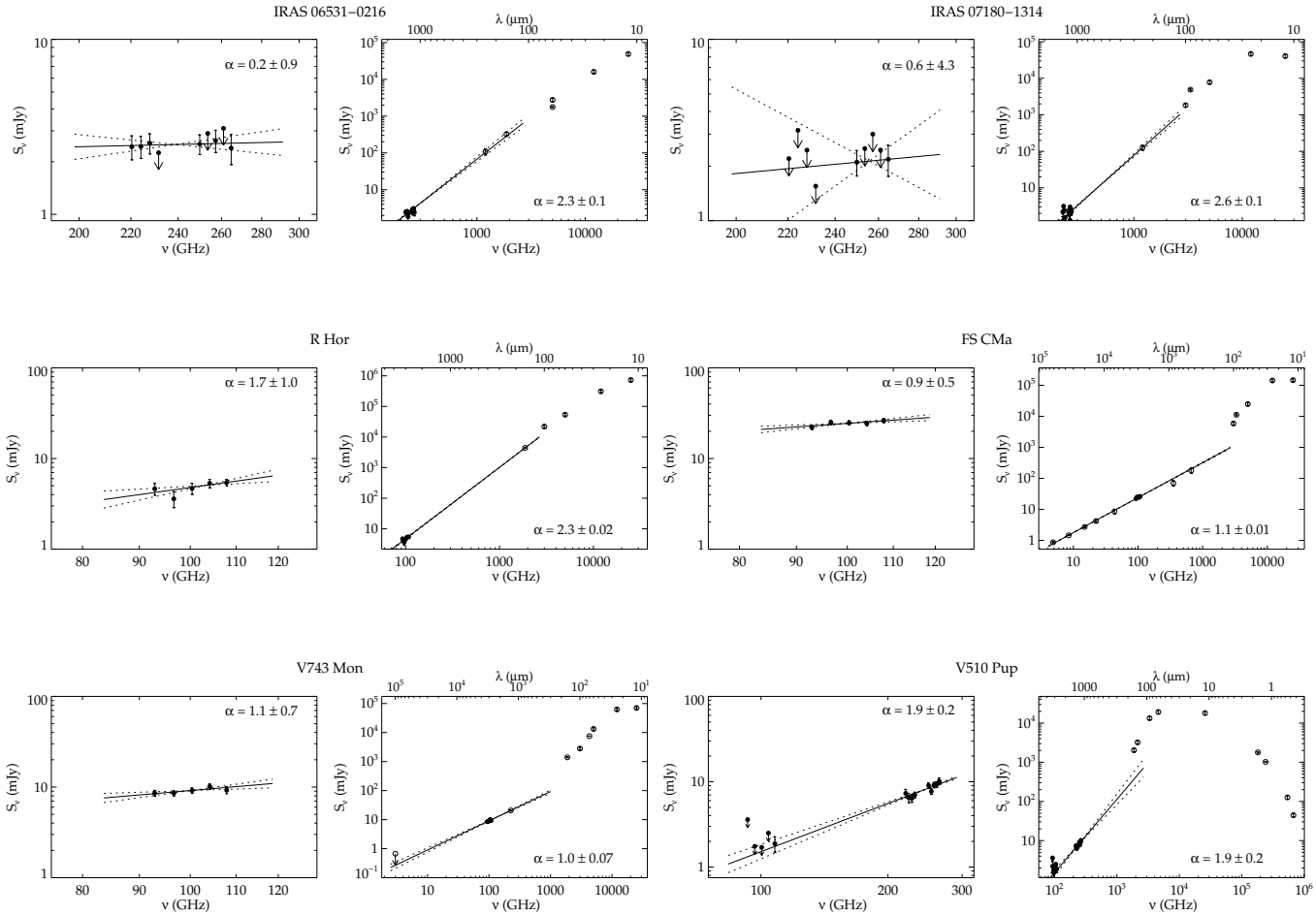

Figure 1. Continuum spectra of targets with positive spectral index. A close-up on the ACA data is displayed on the left-hand side and the whole mm/mid-IR SED on the right.

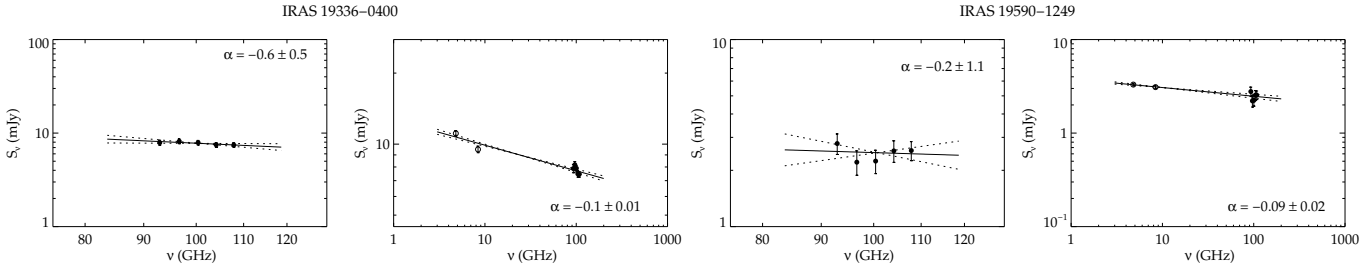

Figure 2. Continuum spectra of sources with negative spectral indices. For each source, the ACA data points are displayed on the left-hand side with their fit, while on the right the result of the extension of the fit to low frequencies is shown.

As can be seen in Figure 1, the values of spectral indices derived by fitting our mm points have large errors. To properly derive reliable spectral indices, we would need much higher signal-to-noise ratio in our data and/or data points sampling better the whole mm/far-IR range. Such an example is V510 Pup, which was detected in both band 3 and 6 and thus has a smaller error on its spectral index.

With the exception of I07180, our targets have spectral indices below 2, even after accounting for their errors. Comparing the far-IR and mm slopes in the SEDs, FS CMa, V743 Mon, and V510 Pup clearly show different distributions: the slope of the far-IR spectrum extrapolated to longer wavelengths appears to be much steeper than the mm distribution. Such a SED is indicative of the presence of two different populations of dust grains and the excess at mm wavelengths is often interpreted as due to large grains grown in a stable circumstellar structure **REFERENCE NATTA/RICCI**. The small spectral index would then be explained as due to a density distribution more complex than that deriving from constant mass loss ($\propto \nu^{-2}$) as well as a possible large size of the grains. It must also be noted that the chemical composition of dust grains in evolved stars is less well known than in the interstellar medium, due to the processes of element enrichment that occur in the atmospheres of these stars.

Unlike the targets above, the spectral indices found in I06531, I07180, and R Hor may match with the extrapolation at longer wavelengths of the far-IR SED, if we consider their errors. For these three stars, we have therefore extended our mm fit to the points at $\lambda > 100$ μm. These extended fits are shown in Figure 1 along with the spectral index derived, which is larger than 2 in all of the three sources. While for I07180 and R Hor the slope of the line fitted after including the sub-mm range is compatible within errors with that derived by fitting only the mm points, this is not the case for I06531. In this latter source, the mm-only spectral index is quite flatter. Since this target was not detected in Band 3, we cannot rule out the presence of a contributions from stellar wind ($S_\nu \propto \nu^{0.6}$), which would be about half as bright in the lower band than in Band 6, thus falling below our detection limit. The presence of emission from a stellar wind within our beam could flatten the SED at long wavelengths.

In IRAS 19336-0400 and 19590-1249, we can confirm that their continuum spectrum is flat with index of $\sim$-0.1, as expected for an ionized shell, confirming previous findings (Cerrigone et al. 2011).

For the sources where we were able to derive a spectral index and dust emissivity, we calculated the mass of dust according to the formulae in Beuther et al. (2005), assuming the distances derived from the Gaia parallaxes and a dust temperature of 100 K. For IRAS 06531-0216, 07180-1314, FS CMa, R Hor, V510 Pup, and V743 Mon, we find dust masses of 10^{-4}, 1.5×10^{-4}, 1.7×10^{-3}, 9×10^{-4}, 3×10^{-4}, and 9×10^{-4} $M_\odot$, respectively. Given the uncertainties related to the distances and the assumed dust temperature and density distribution, these values are reported as an indication of order of magnitude.

COMMENT ON NON-Detections

3.2. *Spectral-line detections*

3.2.1. *IRAS 06531-0216*

Besides CO and ^{13}CO, this source is classified as a C star and displays lines typical of C-rich envelopes, such as HCN, HC_3N, HC_7N, HNC, CS, SiC_2, and the CN and CCH radicals. SiS and SiO (and its isotopomers) are also detected. SiS is detected when $\dot{M}$ is larger than 10^{-6} $M_\odot$ yr^{-1} according to Massalkhi et al, 2019, A&A 628, A62. Clear signs of a fast expansion are seen in several transitions (for example, in CCH and ^{13}CO), pointing to an expansion velocity of ∼30–40 km s^{-1}, which is larger than the typical AGB values.

3.2.2. *IRAS 07145-1428*

This is a C star and we detect lines of CO, CS, HCN, HC_3N, CN, CCH, but also SiO, which is not typical of C-rich envelopes. The CCH radical has a width at zero emission of about 70 km s^{-1}, therefore arising in a region expanding at ∼35 km s^{-1}. The shape of the SiO lines seems to match this expansion velocity as well. This latter finding implies that SiO is arising in regions where the wind is accelerating.

3.2.3. *IRAS 07180-1314*

In Band 3, we detect only CO, SiO (v=0 and v=1), and SO lines. In Band 6, besides these molecules, we also detect SiS and SO_2, HCN, and $H^{13}CN$. A line around 255.04 GHz is tentatively attributed to HCCN, since HCN and $H^{13}CN$ are detected. Finally, a water line is seen around 268.1 GHz. Being classified as an OH/IR star, the detection of O-bearing molecules is expected, while the presence of compounds containing carbon must be related to non-thermal equilibrium chemistry due to shock propagation in the inner regions of the wind (Van de Sande et al. 2018).

3.2.4. *IRAS 08500-3254*

CO, SiO, and HCO+ are detected in Band 3, compatibly to its nature as an M star. Transitions from the same molecules are detected in Band 6. The presence of HCO+ is indicative of a molecular outflow expanding at ∼20 km s^{-1}. HCN and CN are also detected. SiO is detected in several transitions and isotopomers, including lines from vibrationally excited states.

3.2.5. *R Hor*

This target was observed only in Band 3 and we detected CO and ^{13}CO as well as SiO lines from v=0 and v=1 states. The lines detected match with the O-rich nature of the star.

3.2.6. *RV Aqr*

Lines typical of C-rich envelopes such as HCN, HC_3N, CS, CN are detected. In particular, this source is richer than the others in transitions of the CN radical. ^{13}CO has a zero-emission line width of ∼60 km s^{-1}, pointing to an expansion velocity of the envelope around 30 km s^{-1}. The HCN and SiO lines have a clearly triangular shape, indicative of outflowing material, spanning about the same velocity width as ^{13}CO.

3.2.7. *RZ Sgr*

This target was observed only in Band 3 and displays CO, ^{13}CO, SiO (v=0 and v=1), and an HCN line at 88.64 GHz. As expected for its nature as a S star, both C- and O-bearing molecues are seen in its spectrum.

3.2.8. *UU For*

Like RZ Sgr, UU For was observed only in Band 3 and it displays features of CO and ^{13}CO, SO, and SiO (v=0 and v=1), including the isotopomer ^{29}SiO. HCN is also detected, which is somehow surprising, since this is classified as an M star.

3.2.9. *V510 Pup*

This is the source displaying the richest collection of spectral lines in the sample, although in Band 3 only three lines are detected. These lines belong to SiO (tentative detection), CO, and HCN. The CO(1–0) line has a triangular shape and the HCN line has a double peak confirmed in the HCN transitions in Band 6.

In the higher-frequency band, V510 Pup has bright wide CO and ^{13}CO lines. The ^{13}CO line clearly shows an underlying high-velocity component, extending roughly from ~ -160 to ∼255 km s^{-1}, corresponding to an outflow

velocity of ∼200 km s^{-1}. On top of this wide component, there is a central peak at ∼50 km s^{-1}, with two peaks at ∼10 km s^{-1} and 105 km s^{-1}. The width of the high-velocity component matches with that of the CO(2–1) line. Wide shapes are also detected in the features of SiO and its isotopomers, although corresponding to a smaller outflow velocity (∼130 km s^{-1}), which can be interpreted as SiO being present in the gas phase in inner regions, where the outflow is still accelerating. HCN, HC_3N, HNC, and CS share the same shape with three peaks seen in the bright CO components. Fetaures of the CCH radical are also detected around 262 GHz, with overlapping pairs of peaks. Each pair is indicated by arrows in Figure 28 and the peaks in every pair are separated by ∼95 km s^{-1} from each other.

This object hosts a central binary and the molecular high-velocity outflow seen in many transitions in our survey is seen also in Hα (Manick et al. 2021).

3.3. *HCN to HNC ratio*

The HCN to HNC ratio has been investigated in several sources and has been found to be higher in stars still close to the AGB and then decrease in the pPN nd PN stages (schmidt and Ziurys 2017, ApJ, 835, 79). This behavior has been explained as HNC abundance increasing as a consequence of the larger amount of ionizing photons leading to the production of HNC. Stars close to the AGB display values in the range 40–300 (Woods et al. 2003, A&A, 402, 617), while in pPNe this drops below 10 and tends to reach unity in PNe (Schmidt and Ziurys 2017). We detected HNC in 06531 and V510 only in the 2–1 transition at 271 GHz, while the 1–0 line in Band 3 falls below our detection limit. In V510, the ratio is much smaller than in 06531. This is a further indication of the more evolved stage of this source, which has entered the post-AGB phase.

3.4. *^{12}CO to ^{13}CO in Band 6*

According to CHATGPT:

^{12}CO/^{13}CO RATIOS IN DIFFERENT CLASSES OF STARS

The ^{12}CO/^{13}CO ratio varies significantly across different classes of stars, reflecting their distinct nucleosynthetic processes and evolutionary histories. Here are the typical values for various types of stars:

MAIN SEQUENCE STARS

- **Solar-type stars**: The ^{12}CO/^{13}CO ratio in main sequence stars like the Sun is typically around 89, similar to the solar value. This ratio is representative of the initial composition of the interstellar medium from which these stars formed.

RED GIANT BRANCH (RGB) STARS

- **Low- and intermediate-mass stars**: During the red giant phase, the first dredge-up occurs, which brings material from the interior to the surface. This process increases the ^{13}C abundance, resulting in lower ^{12}CO/^{13}CO ratios, typically in the range of 20 to 30.

ASYMPTOTIC GIANT BRANCH (AGB) STARS

- **Carbon-rich AGB stars (Carbon stars)**: These stars have undergone significant third dredge-up events, which bring ^{12}C to the surface. As a result, the ^{12}CO/^{13}CO ratios are higher, often exceeding 40 and can be as high as 90 or more.

- **Oxygen-rich AGB stars**: These stars have lower ^{12}CO/^{13}CO ratios compared to carbon stars, usually ranging from 20 to 40, due to less efficient carbon dredge-up.

SUPERGIANTS

- **Red supergiants**: These massive stars can have a wide range of ^{12}CO/^{13}CO ratios depending on their initial mass and evolutionary stage. Typical values range from 10 to 30.

- **Yellow supergiants**: These stars often show lower ^{12}CO/^{13}CO ratios, around 10 to 20, reflecting extensive internal mixing and nucleosynthesis processes.

Table 5. $^{12}CO/^{13}CO$ ratio.

Target	$^{12}CO/^{13}CO$
IRAS 08500-3254	4.2 ± 0.5^{a}
R Hor	25 ± 2
RV Aqr	23 ± 2
RZ Sgr	5.6 ± 0.4
UU For	16 ± 1
IRAS 06531-0216	58 ± 4
IRAS 07180-1314	3.6 ± 0.3
V510 Pup	8.8 ± 0.6

[a]This is the mean of the values derived from the transitions in bands 3 and 6.

POST-AGB STARS AND PLANETARY NEBULAE

- **Post-AGB stars**: These stars can show a range of $^{12}CO/^{13}CO$ ratios depending on their previous AGB phase. Carbon-rich post-AGB stars may have high ratios (similar to carbon AGB stars), while oxygen-rich post-AGB stars have lower ratios.

- **Planetary nebulae**: The $^{12}CO/^{13}CO$ ratio in planetary nebulae can vary widely, typically reflecting the nucleosynthetic history of the progenitor star. Ratios can range from as low as 10 to as high as 90.

MOLECULAR CLOUDS

- **Interstellar medium (ISM)**: In the ISM, the typical $^{12}CO/^{13}CO$ ratio is around 69, although this can vary depending on the region within the galaxy.

SPECIFIC EXAMPLES

- **IRC+10216 (CW Leo)**: A well-known carbon star with a high $^{12}CO/^{13}CO$ ratio, often around 45 to 60.

- **IK Tau**: An oxygen-rich AGB star with a $^{12}CO/^{13}CO$ ratio around 20 to 30.

- **VY Canis Majoris**: A red supergiant with a $^{12}CO/^{13}CO$ ratio in the range of 15 to 25.

These ratios provide valuable insights into the nucleosynthetic processes and evolutionary stages of different types of stars.

The $^{12}CO/^{13}CO$ ratio varies among C-rich, O-rich, and S-type stars on the Asymptotic Giant Branch (AGB). This ratio serves as an important diagnostic of stellar nucleosynthesis and the evolutionary status of AGB stars.

C-RICH AGB STARS

- C-rich stars have an abundance of carbon over oxygen, leading to the formation of carbon-rich molecules like C_2H_2, CN, and HCN.

- $^{12}CO/^{13}CO$ Ratio: C-rich AGB stars often have a higher $^{12}CO/^{13}CO$ ratio, typically around 20-90. The higher ratio is due to the third dredge-up process, which brings up ^{12}C from the helium-burning shell to the surface, enriching the outer envelope with ^{12}C.

O-RICH AGB STARS

- O-rich stars have more oxygen than carbon, resulting in oxygen-rich molecules like H_2O, SiO, and TiO.
- $^{12}CO/^{13}CO$ Ratio: O-rich AGB stars generally exhibit a lower $^{12}CO/^{13}CO$ ratio compared to C-rich stars, usually around 10-30. The lower ratio reflects the initial abundance of ^{13}C and the effects of the first dredge-up and possibly the second dredge-up, which increase the ^{13}C content relative to ^{12}C.

S-TYPE AGB STARS

- S-type stars are a transitional type between O-rich and C-rich stars, characterized by a roughly equal abundance of carbon and oxygen. This balance leads to the formation of ZrO and other s-process elements.
- $^{12}CO/^{13}CO$ Ratio: S-type stars typically have an intermediate $^{12}CO/^{13}CO$ ratio, often falling between the values observed in O-rich and C-rich stars, generally around 15-50. This intermediate ratio reflects the balanced nature of these stars in terms of carbon and oxygen abundances and the nucleosynthetic processes affecting their envelopes.

SUMMARY OF $^{12}CO/^{13}CO$ *RATIOS*

- **C-rich AGB Stars**: $^{12}CO/^{13}CO \approx 20 - 90$
- **O-rich AGB Stars**: $^{12}CO/^{13}CO \approx 10 - 30$
- **S-type AGB Stars**: $^{12}CO/^{13}CO \approx 15 - 50$

These ratios are influenced by the initial stellar composition and the internal nucleosynthetic processes, including dredge-ups, that bring carbon and other elements to the stellar surface during the AGB phase.

APPENDIX

ATLAS OF THE SPECTRAL LINES DETECTED IN THE SURVEY

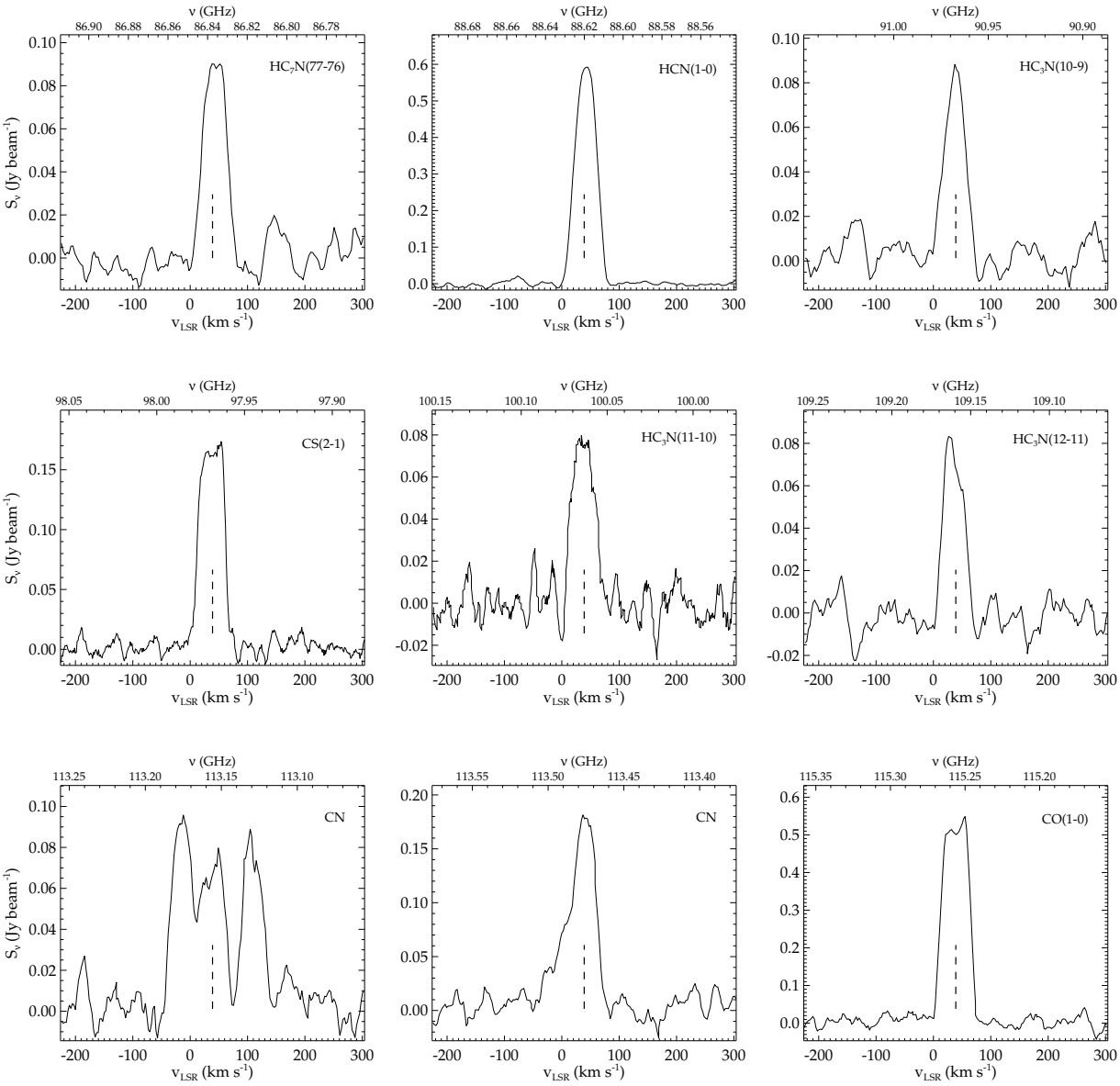

Figure 3. Spectral lines detected towards IRAS 06531 − 0216 at 3 mm. The dashed vertical line indicates the stellar velocity.

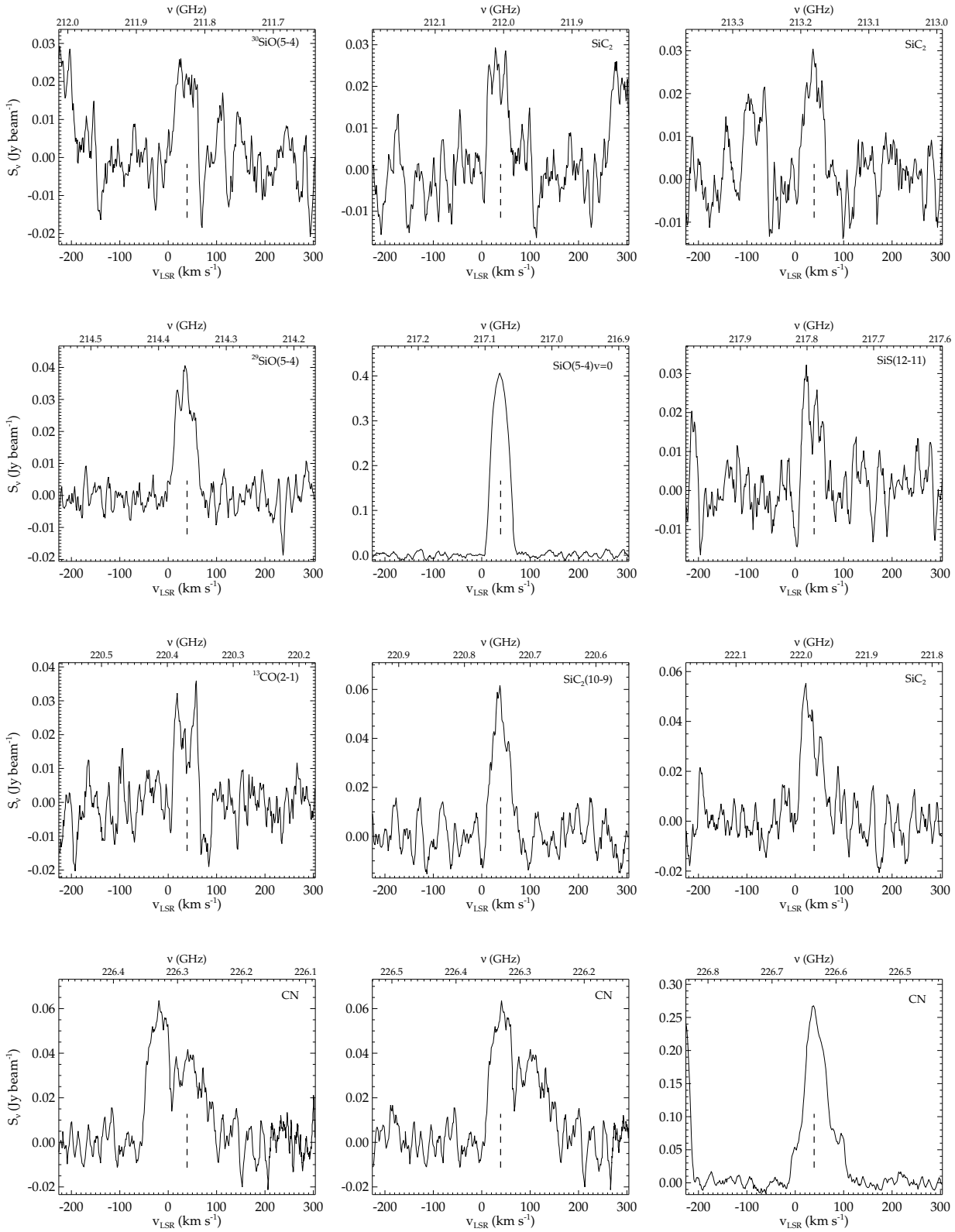

Figure 4. Spectral lines detected towards IRAS 06531 − 0216 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

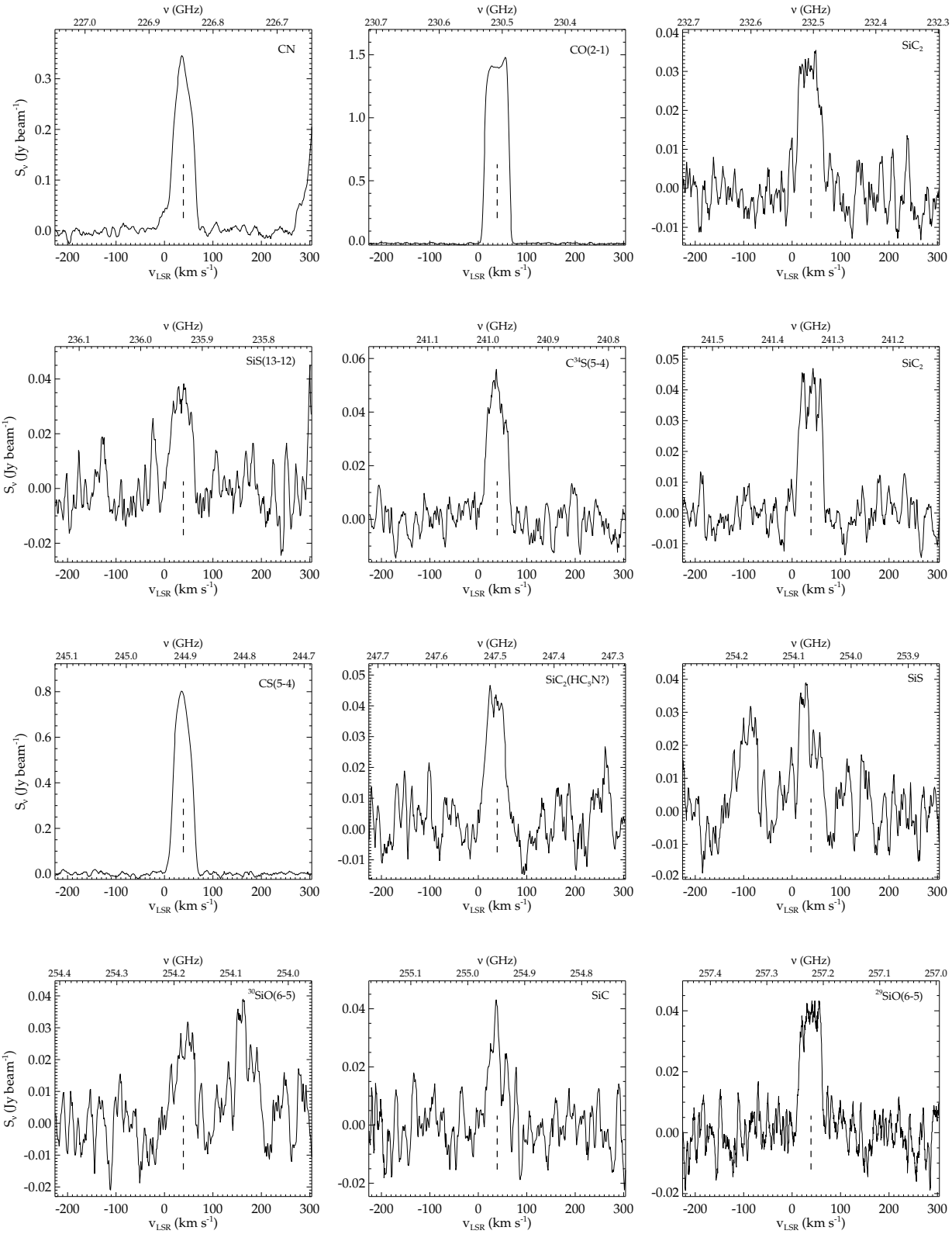

Figure 5. Spectral lines detected towards IRAS 06531 − 0216 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

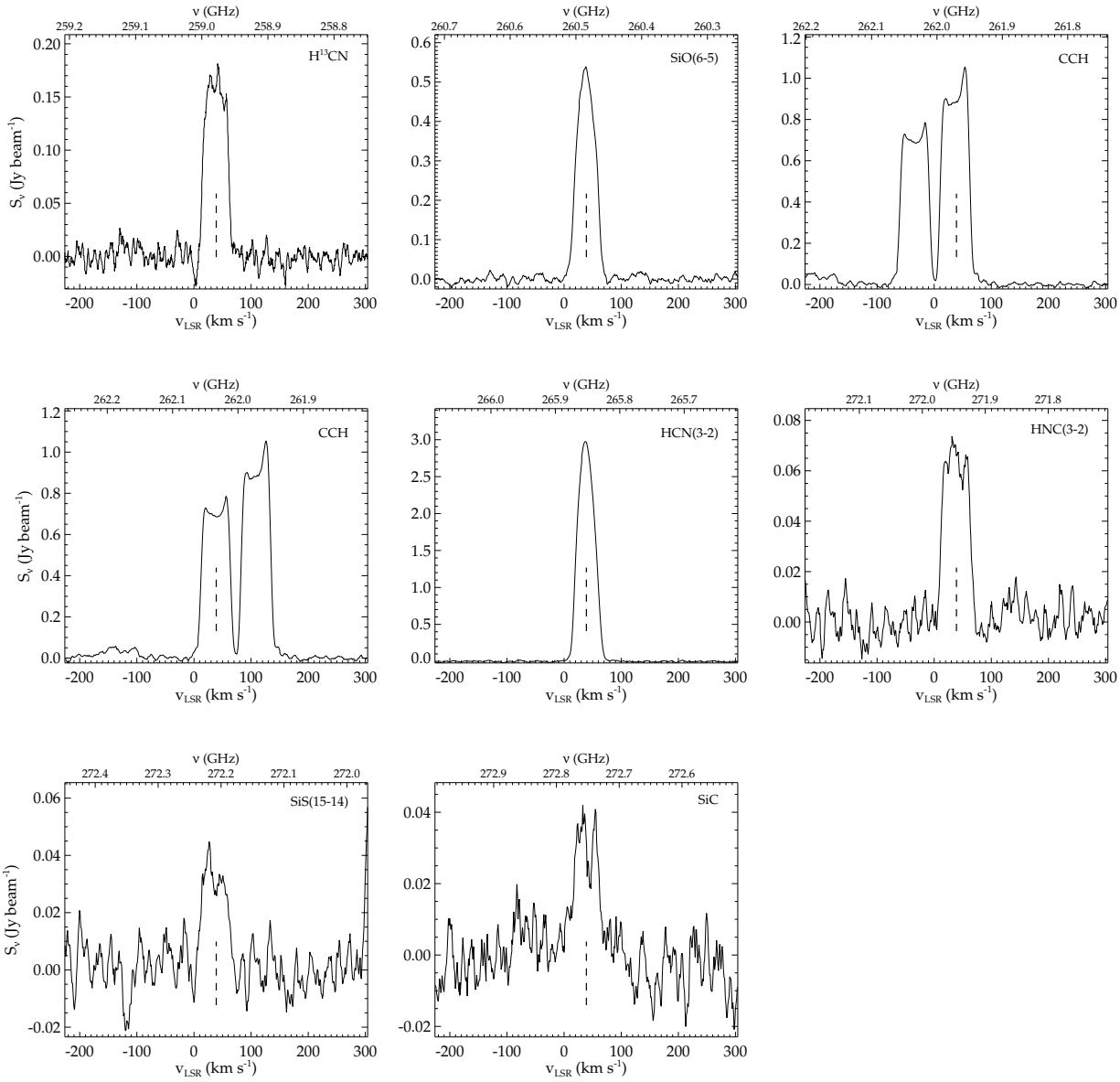

Figure 6. Spectral lines detected towards IRAS 06531 − 0216 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

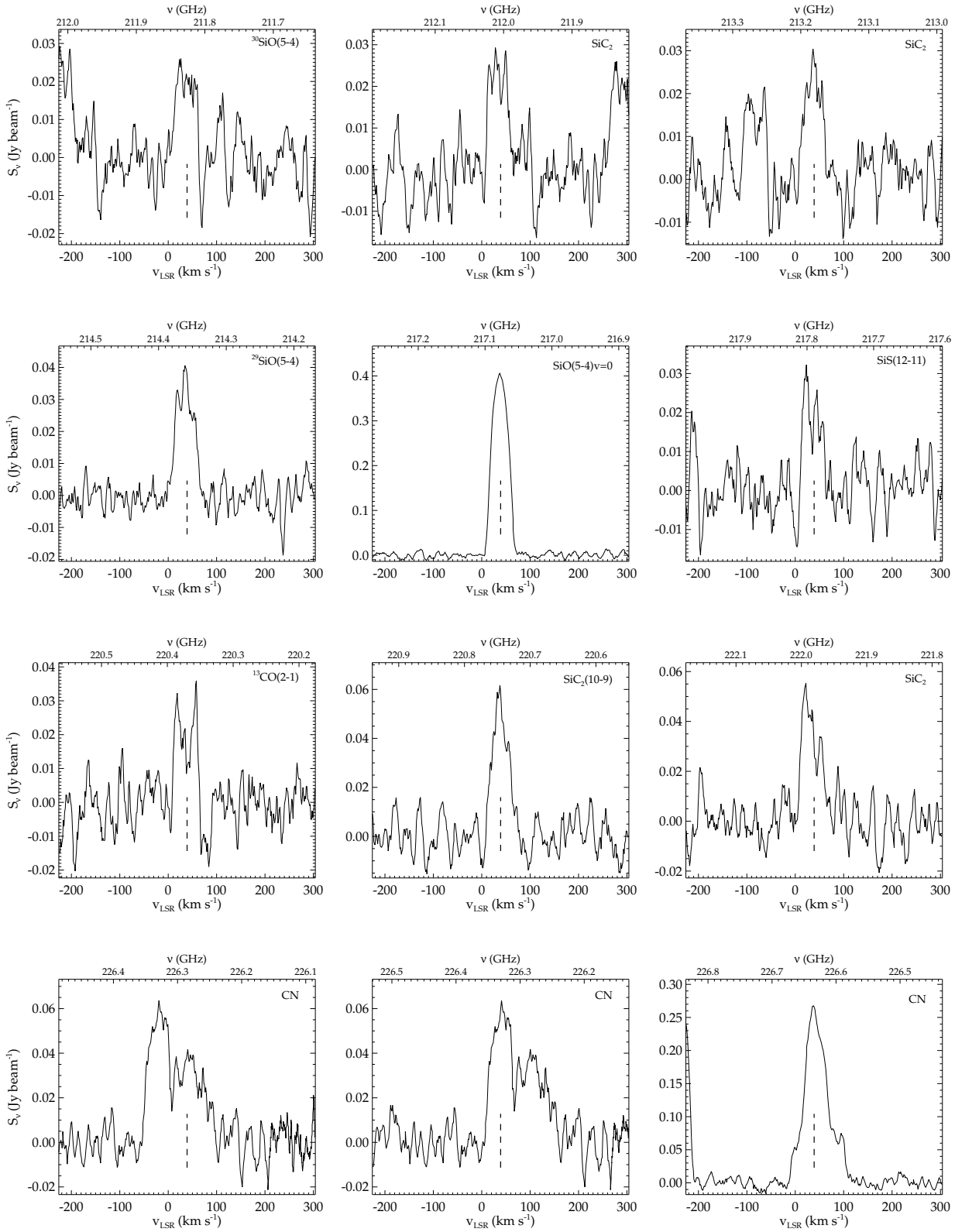

Figure 7. Spectral lines detected towards IRAS 06531 − 0216 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

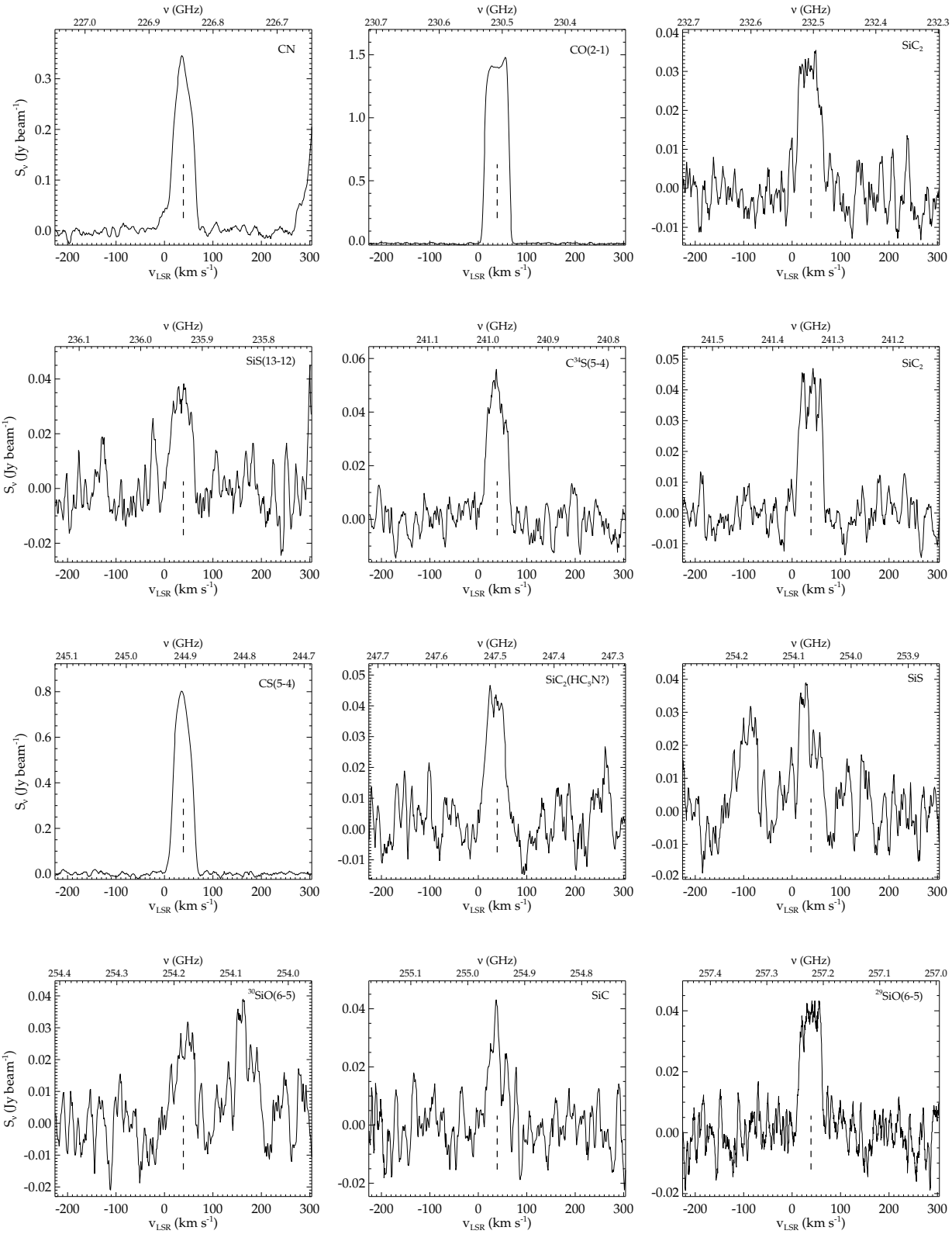

Figure 8. Spectral lines detected towards IRAS 06531 − 0216 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

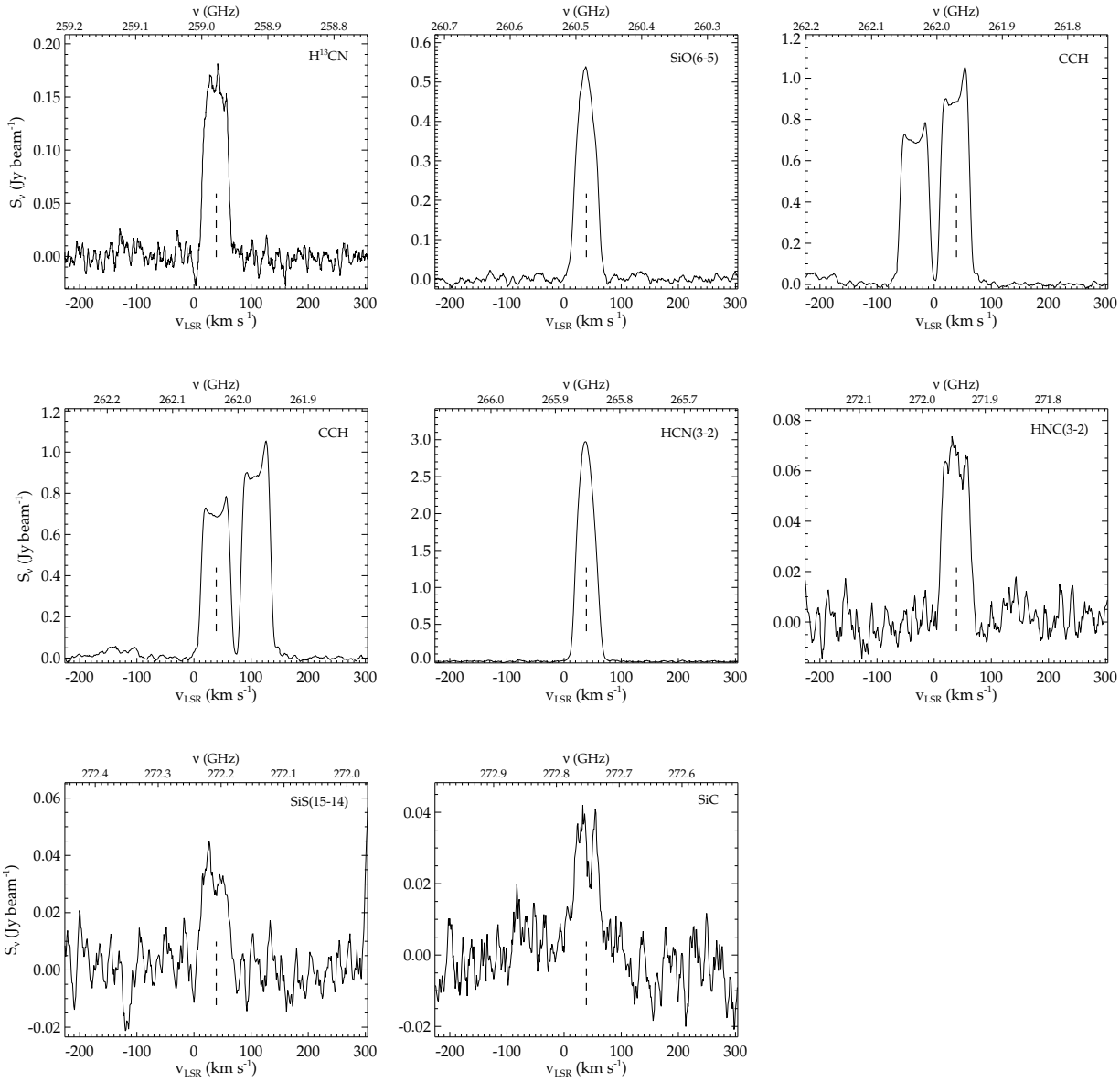

Figure 9. Spectral lines detected towards IRAS 06531 − 0216 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

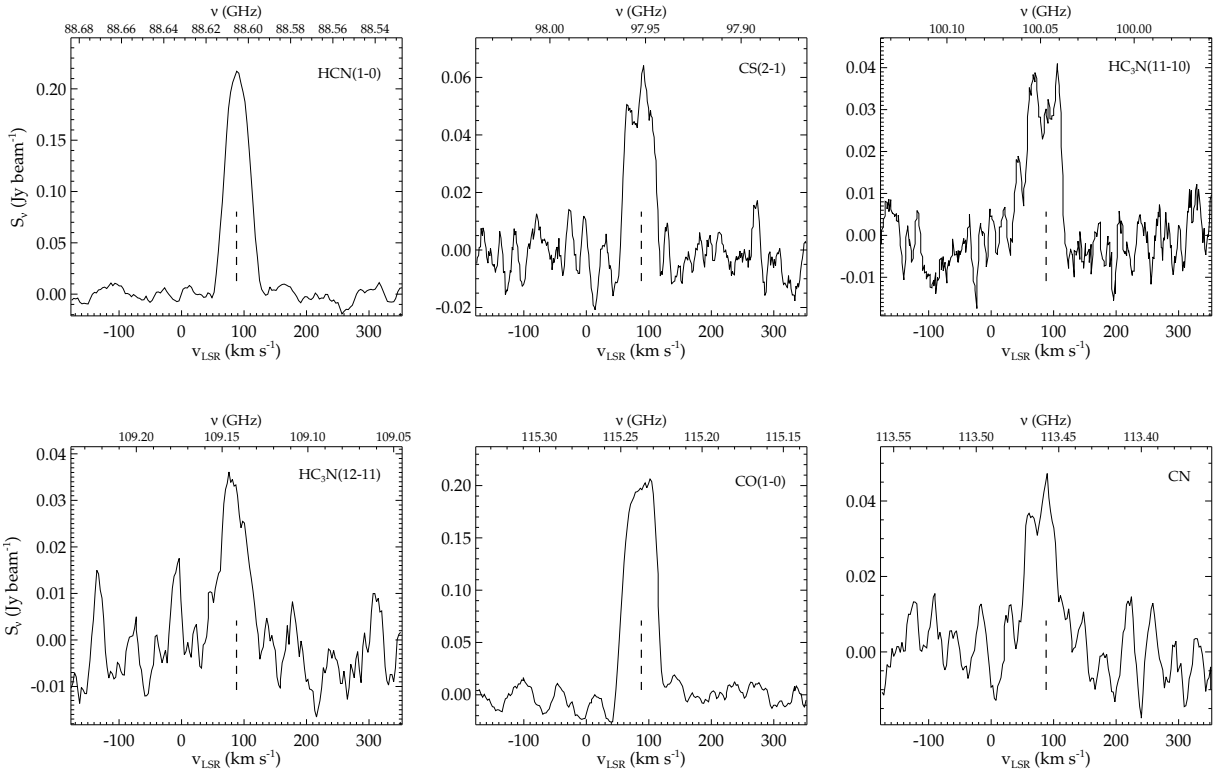

Figure 10. Spectral lines detected towards IRAS 07145 − 1428 at 3 mm. The dashed vertical line indicates the stellar velocity.

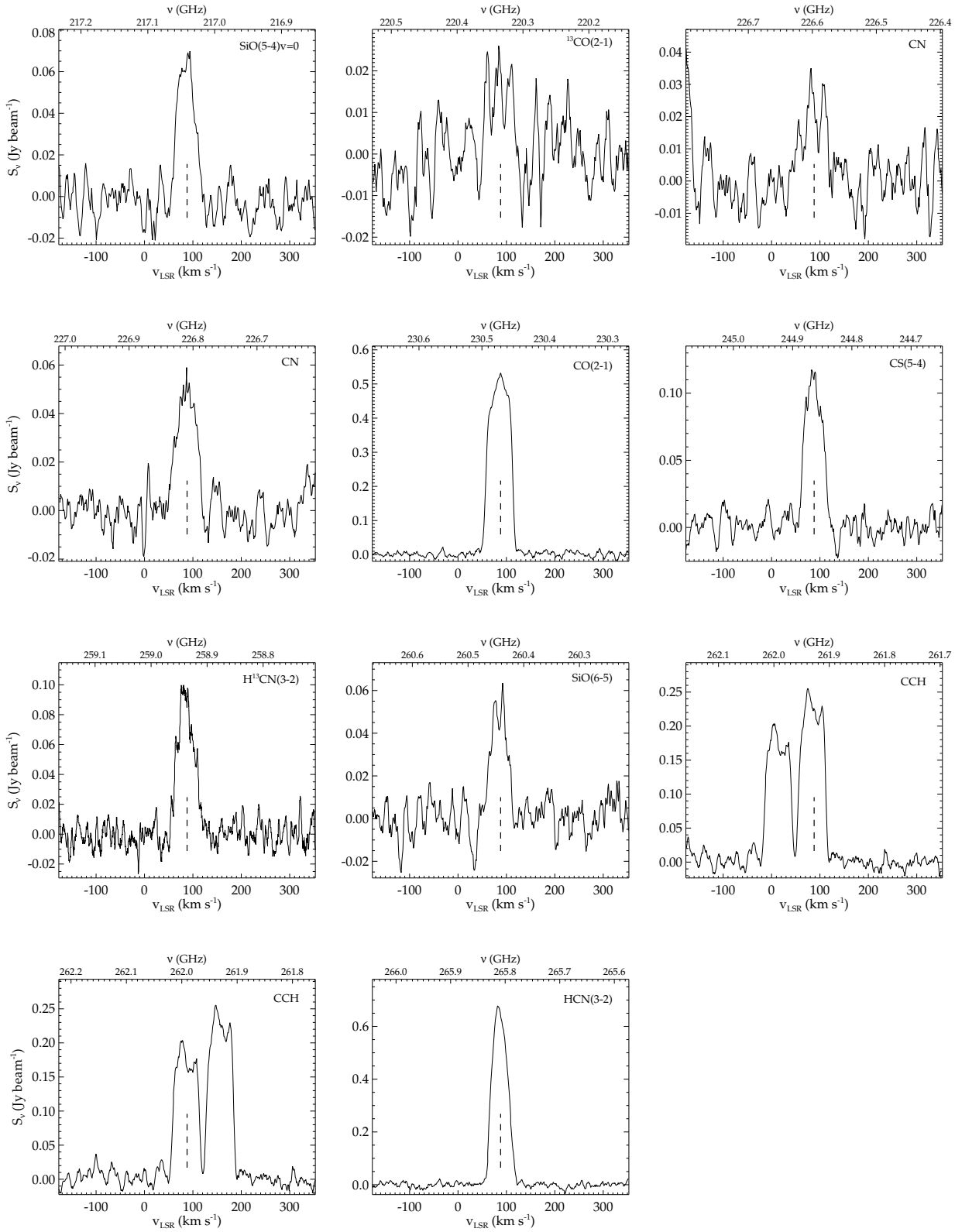

Figure 11. Spectral lines detected towards IRAS 07145 − 1428 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

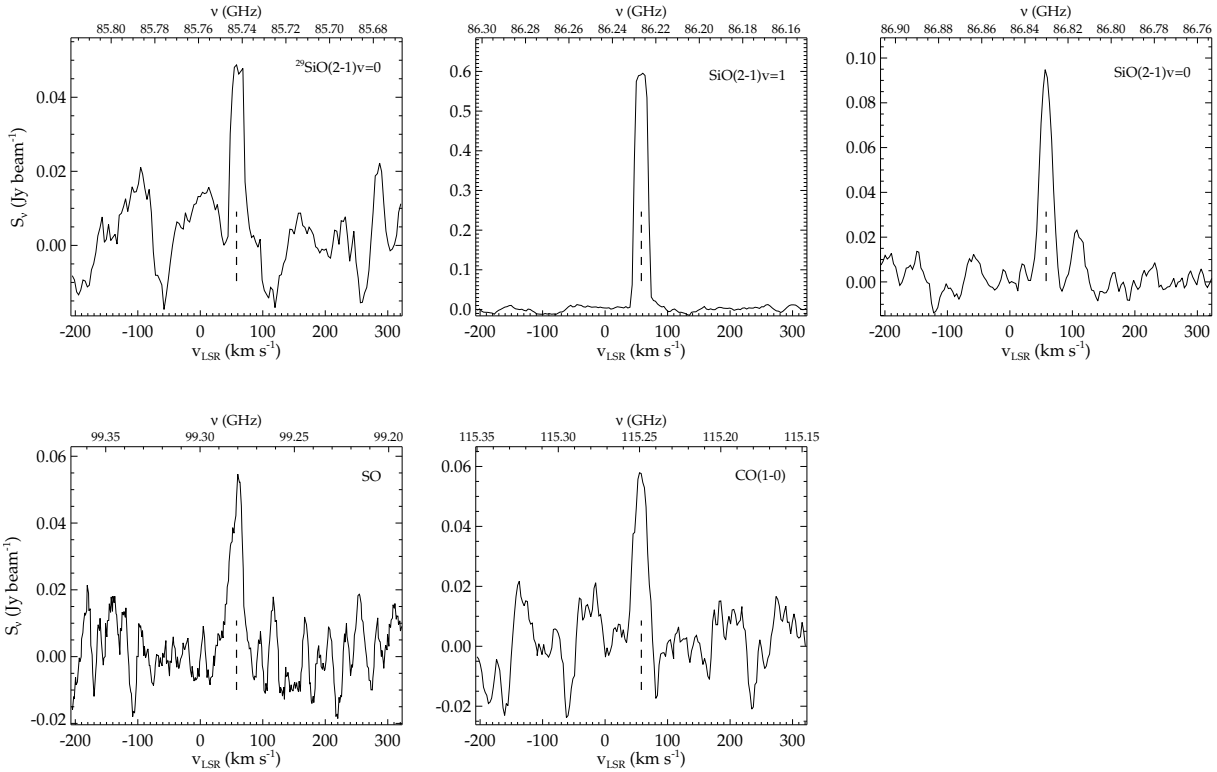

Figure 12. Spectral lines detected towards IRAS 07180 − 1314 at 3 mm. The dashed vertical line indicates the stellar velocity.

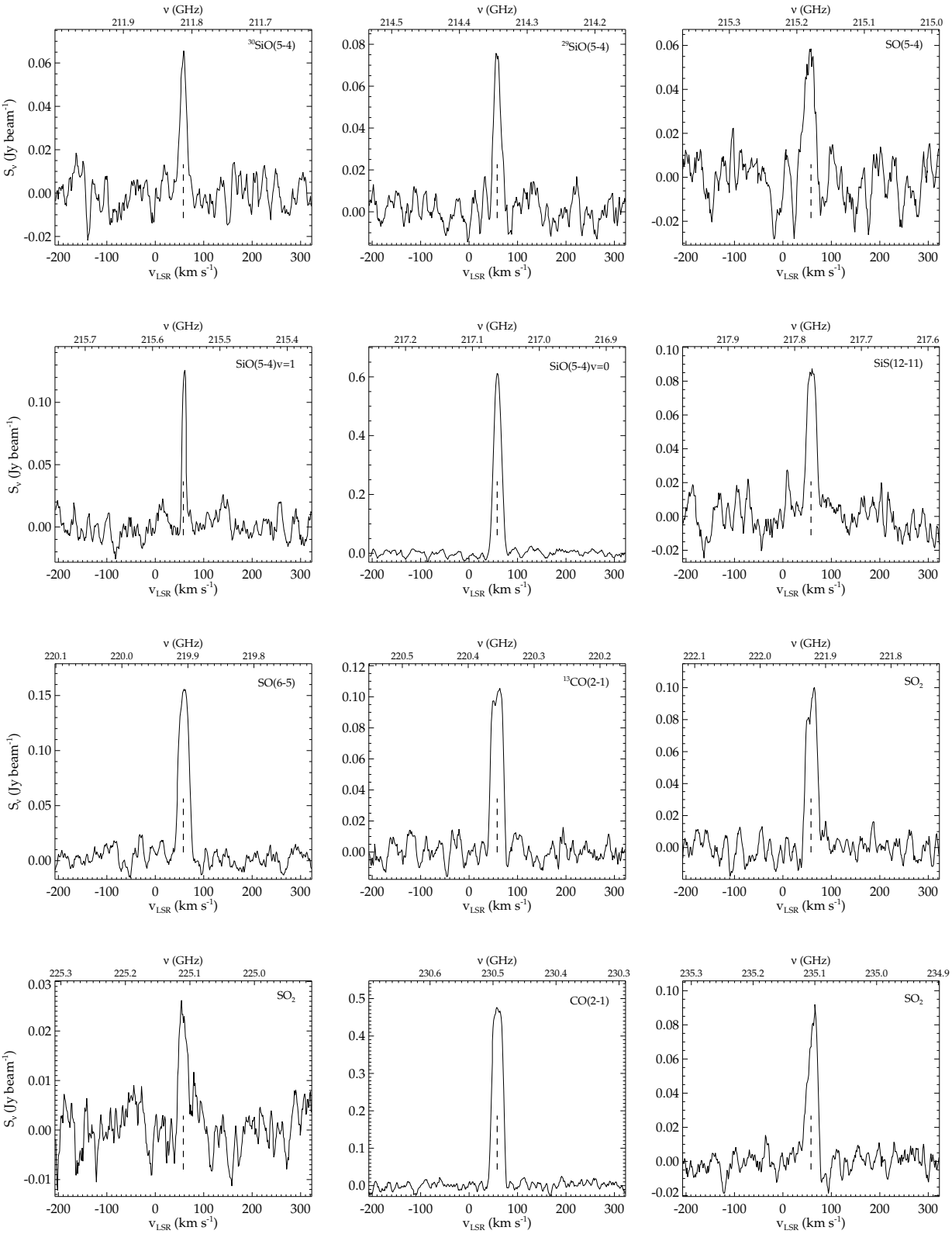

Figure 13. Spectral lines detected towards IRAS 07180−1314 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

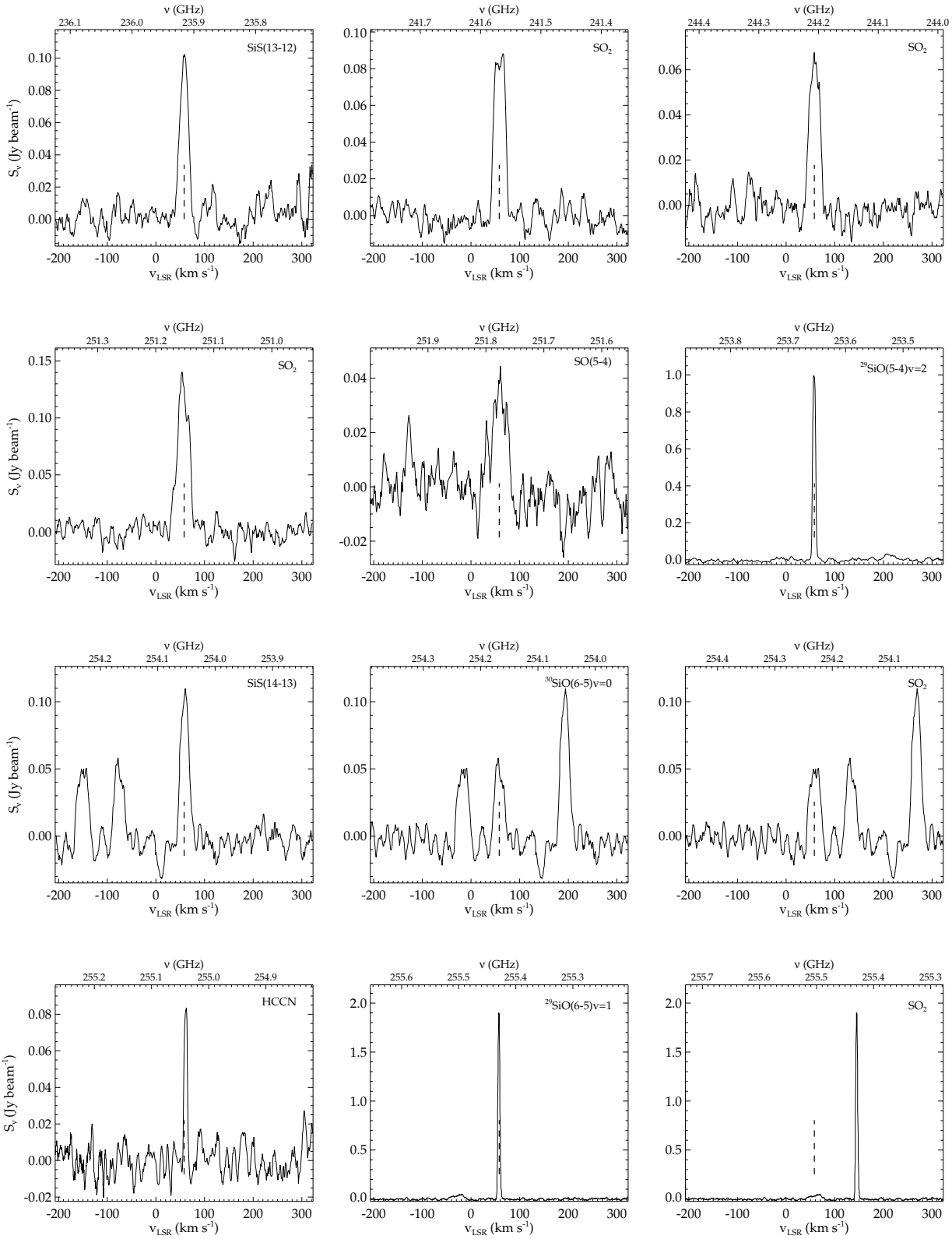

Figure 14. Spectral lines detected towards IRAS 07180 − 1314 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

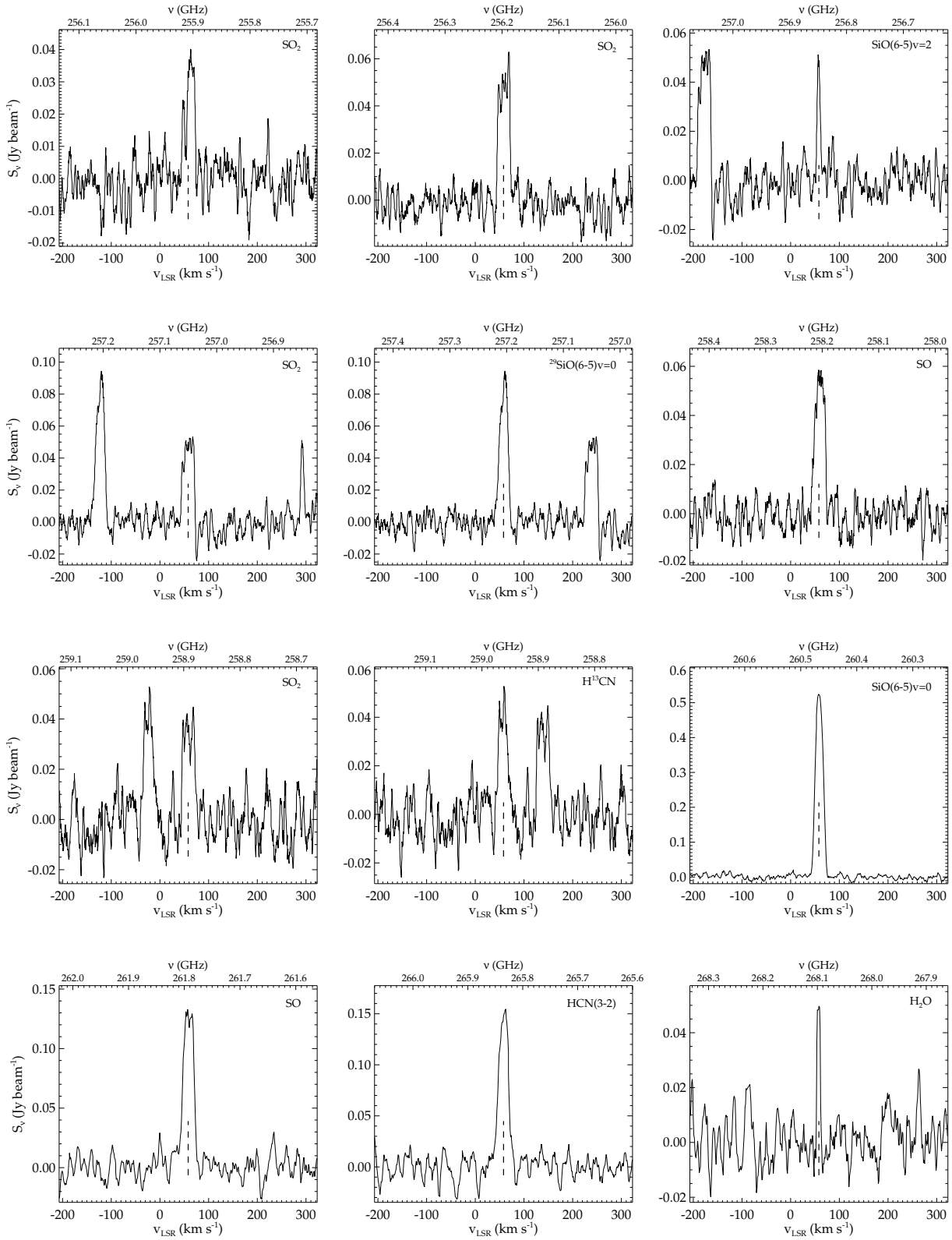

Figure 15. Spectral lines detected towards IRAS 07180 − 1314 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

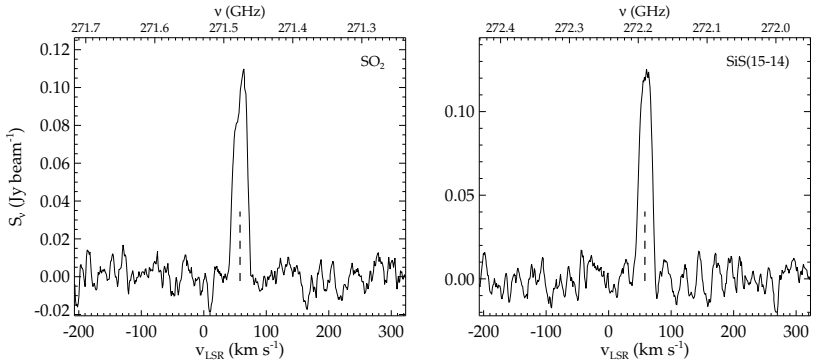

Figure 16. Spectral lines detected towards IRAS 07180 − 1314 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

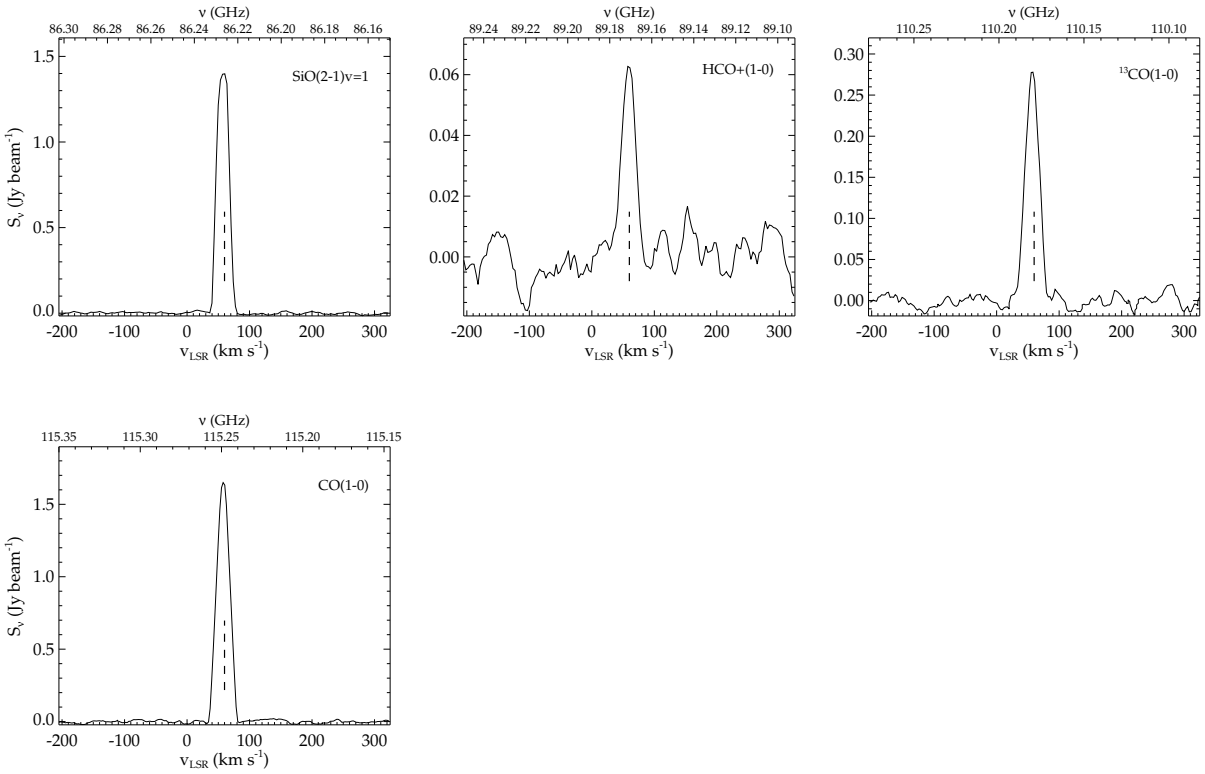

Figure 17. Spectral lines detected towards IRAS 08500 − 3254 at 3 mm. The dashed vertical line indicates the stellar velocity.

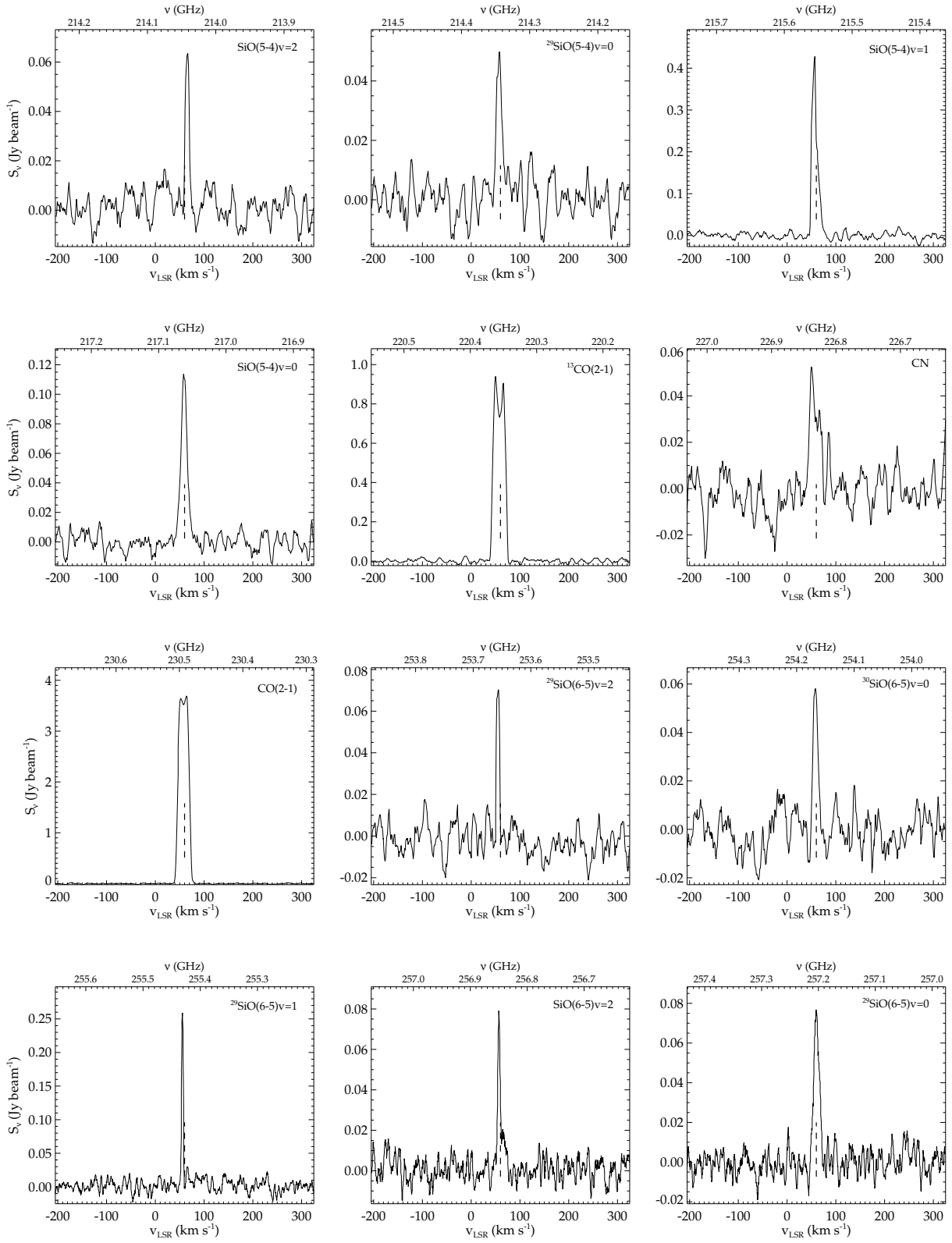

Figure 18. Spectral lines detected towards IRAS 08500 − 3254 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

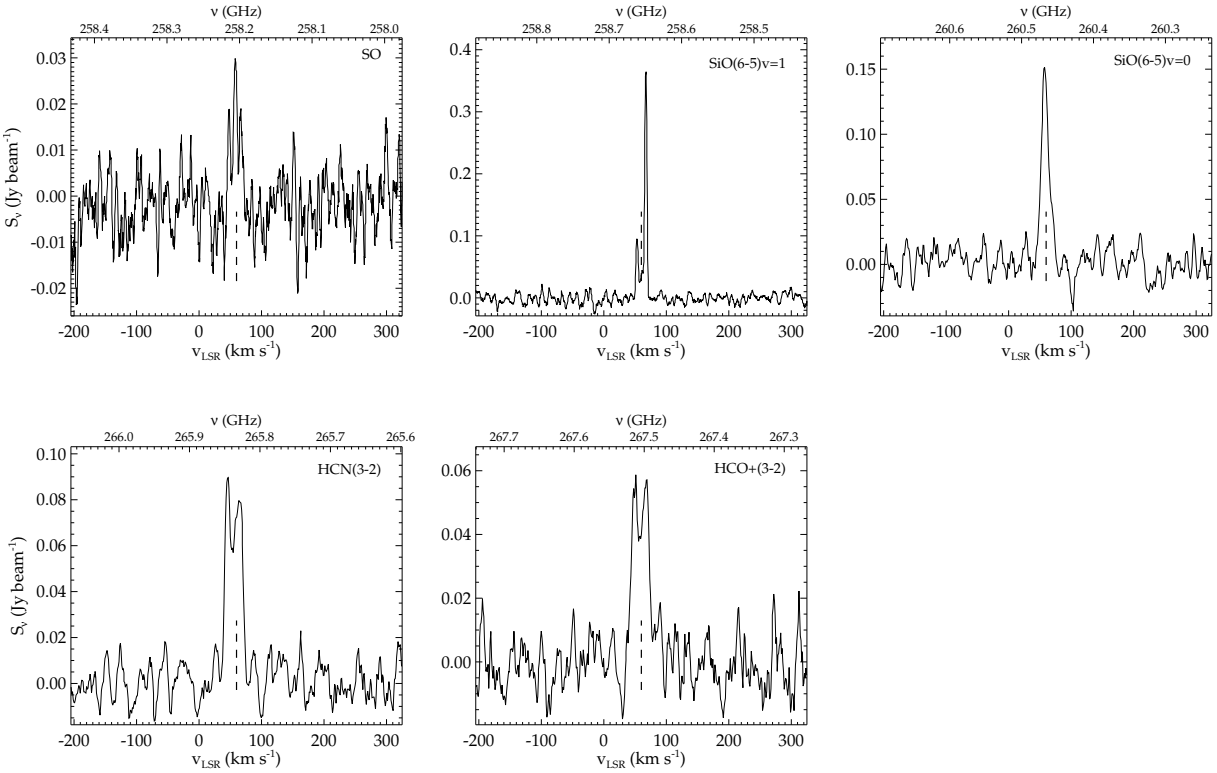

Figure 19. Spectral lines detected towards IRAS 08500 − 3254 at 1.3 mm. The dashed vertical line indicates the stellar velocity.

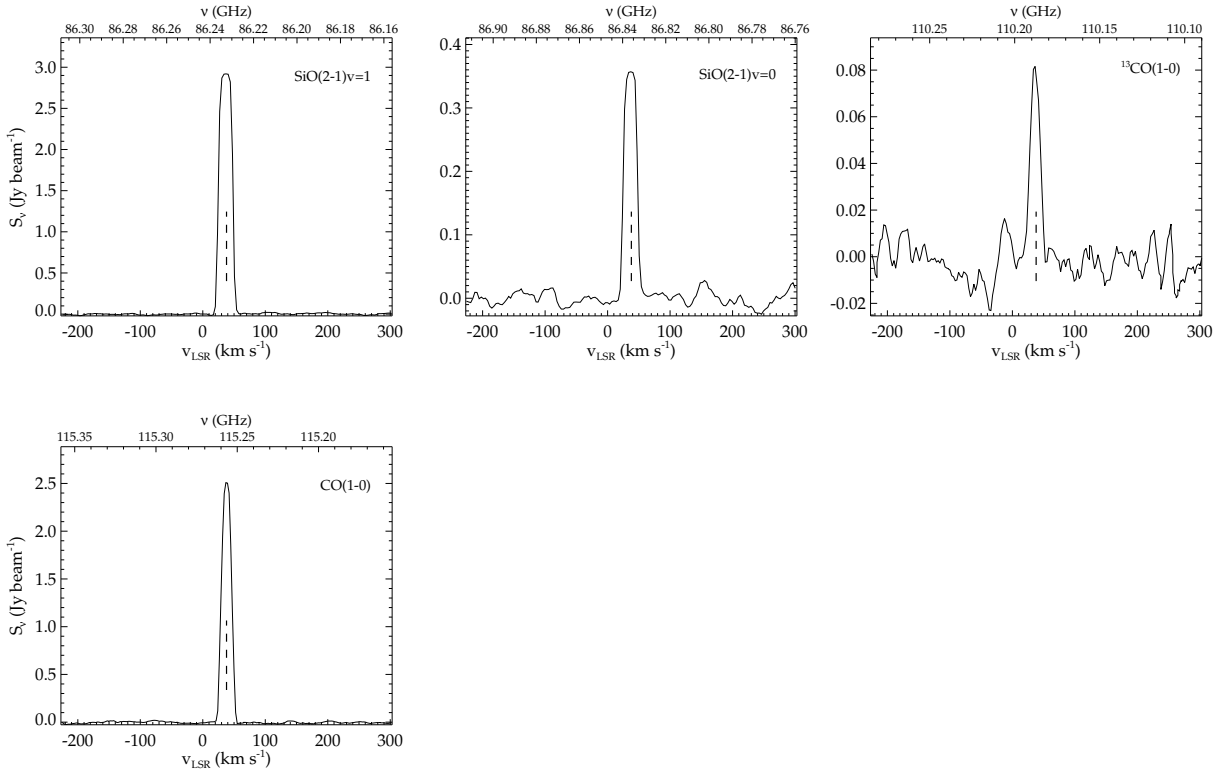

Figure 20. Spectral lines detected towards R Hor at 3 mm. The dashed vertical line indicates the stellar velocity.

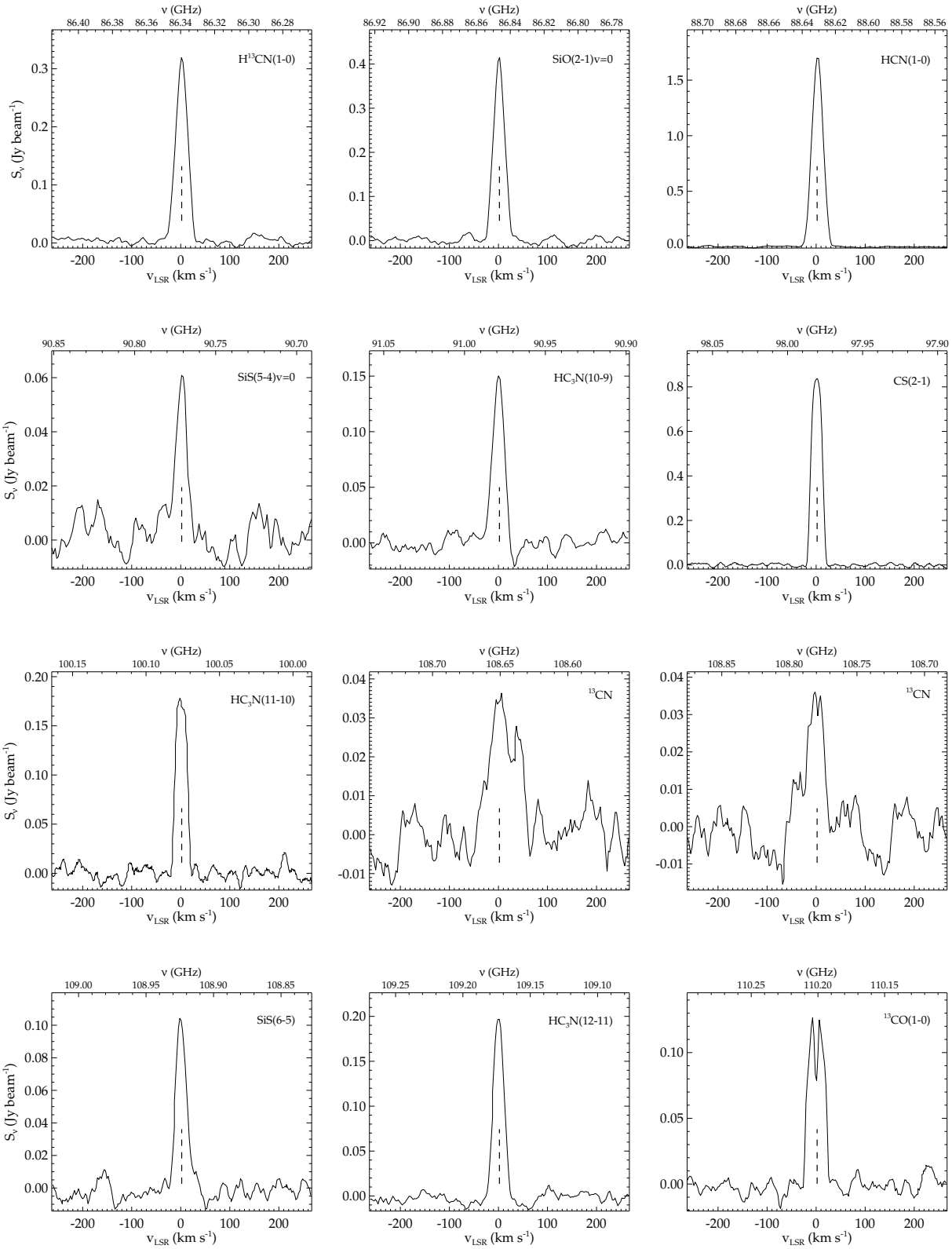

Figure 21. Spectral lines detected towards RV Aqr at 3 mm. The dashed vertical line indicates the stellar velocity.

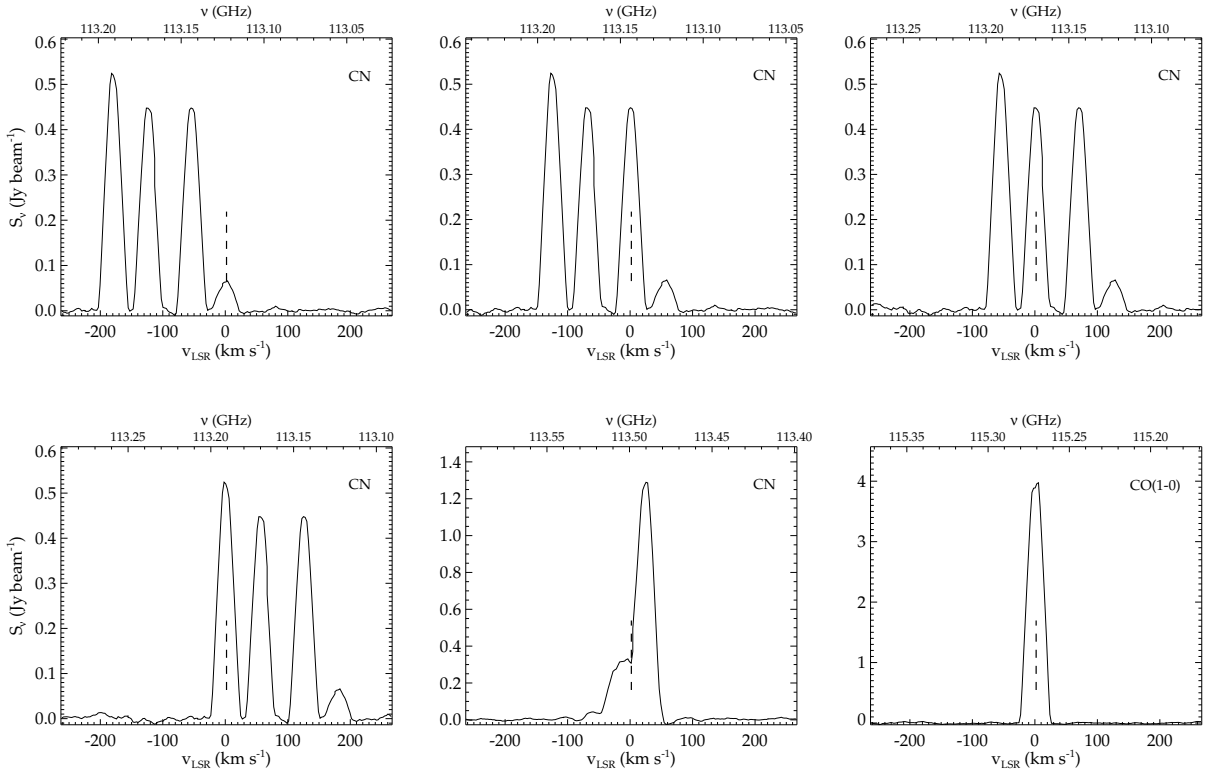

Figure 22. Spectral lines detected towards RV Aqr at 3 mm. The dashed vertical line indicates the stellar velocity.

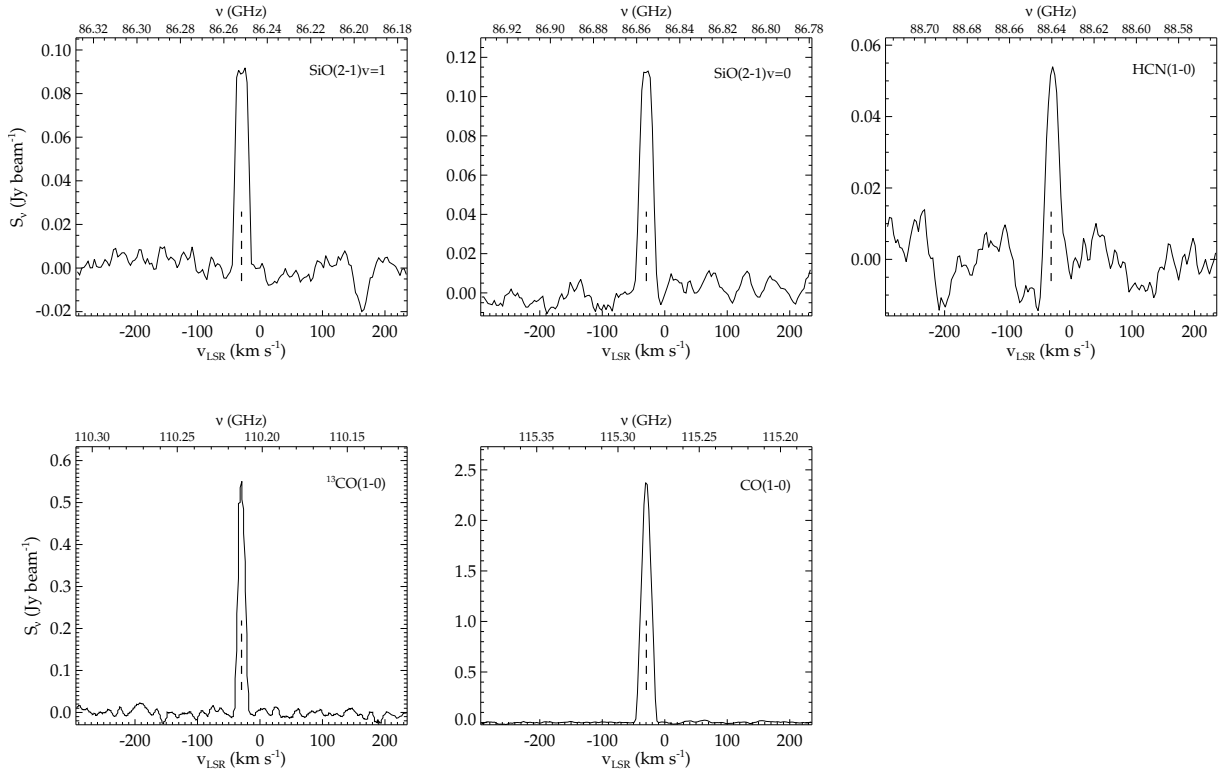

Figure 23. Spectral lines detected towards RZ Sgr at 3 mm. The dashed vertical line indicates the stellar velocity.

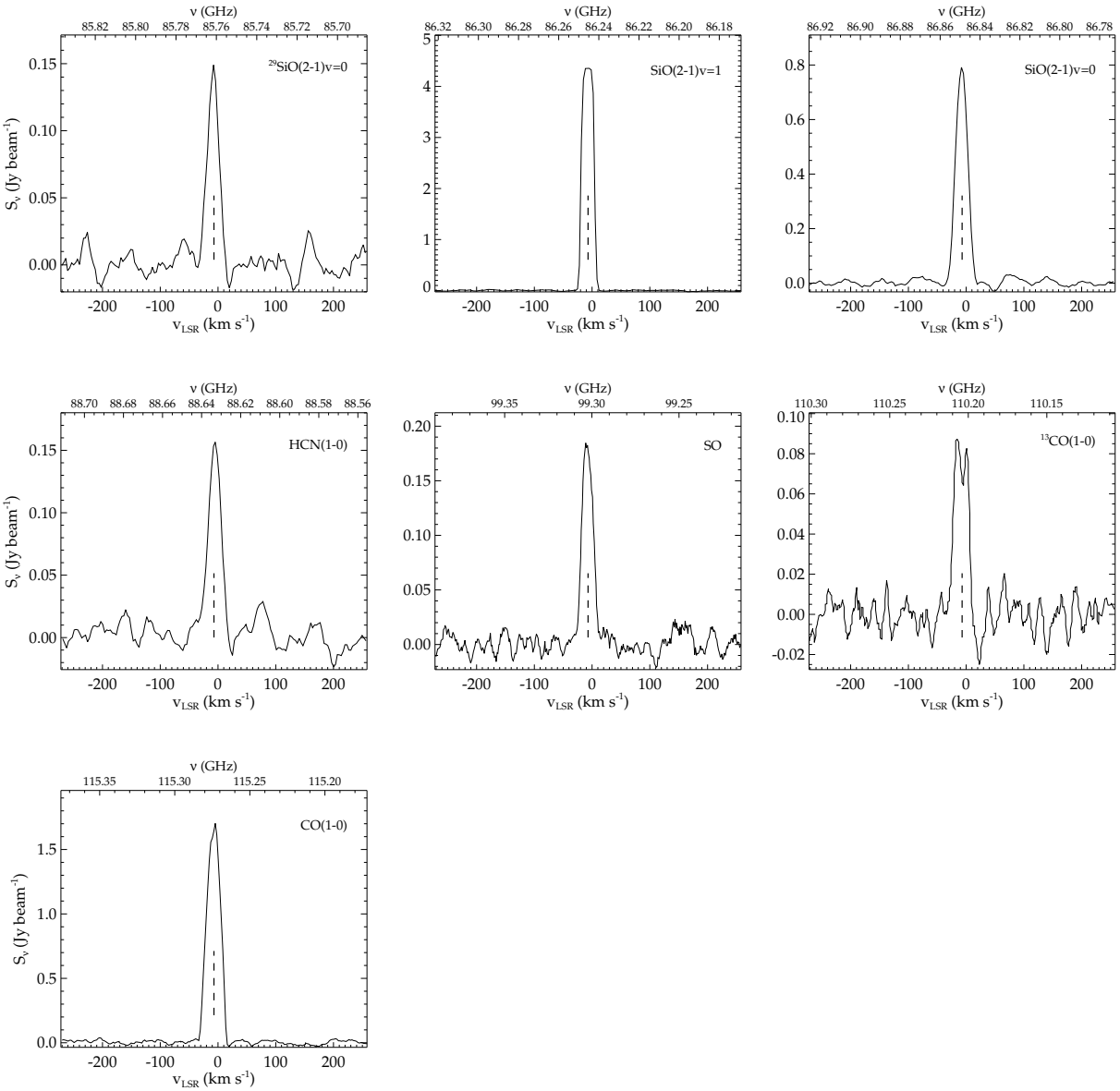

Figure 24. Spectral lines detected towards UU For at 3 mm. The dashed vertical line indicates the stellar velocity.

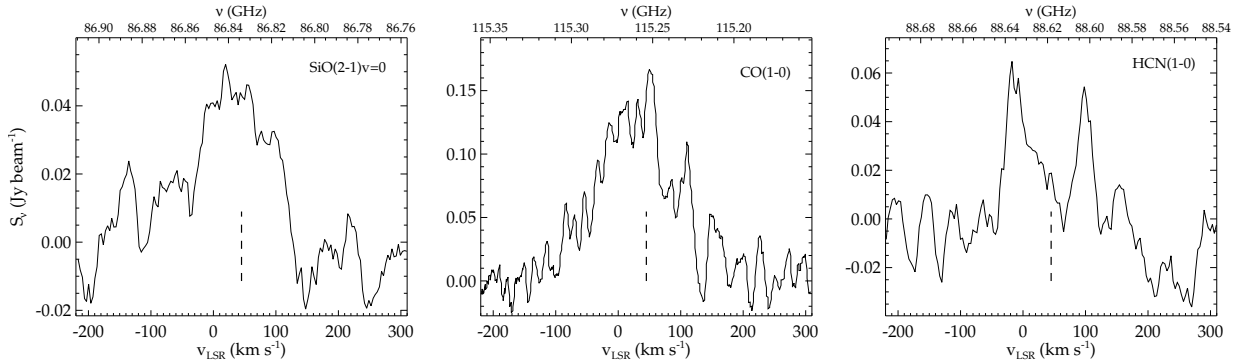

Figure 25. Spectral lines detected towards V510 Pup at 3 mm. The dashed vertical line indicates the stellar velocity.

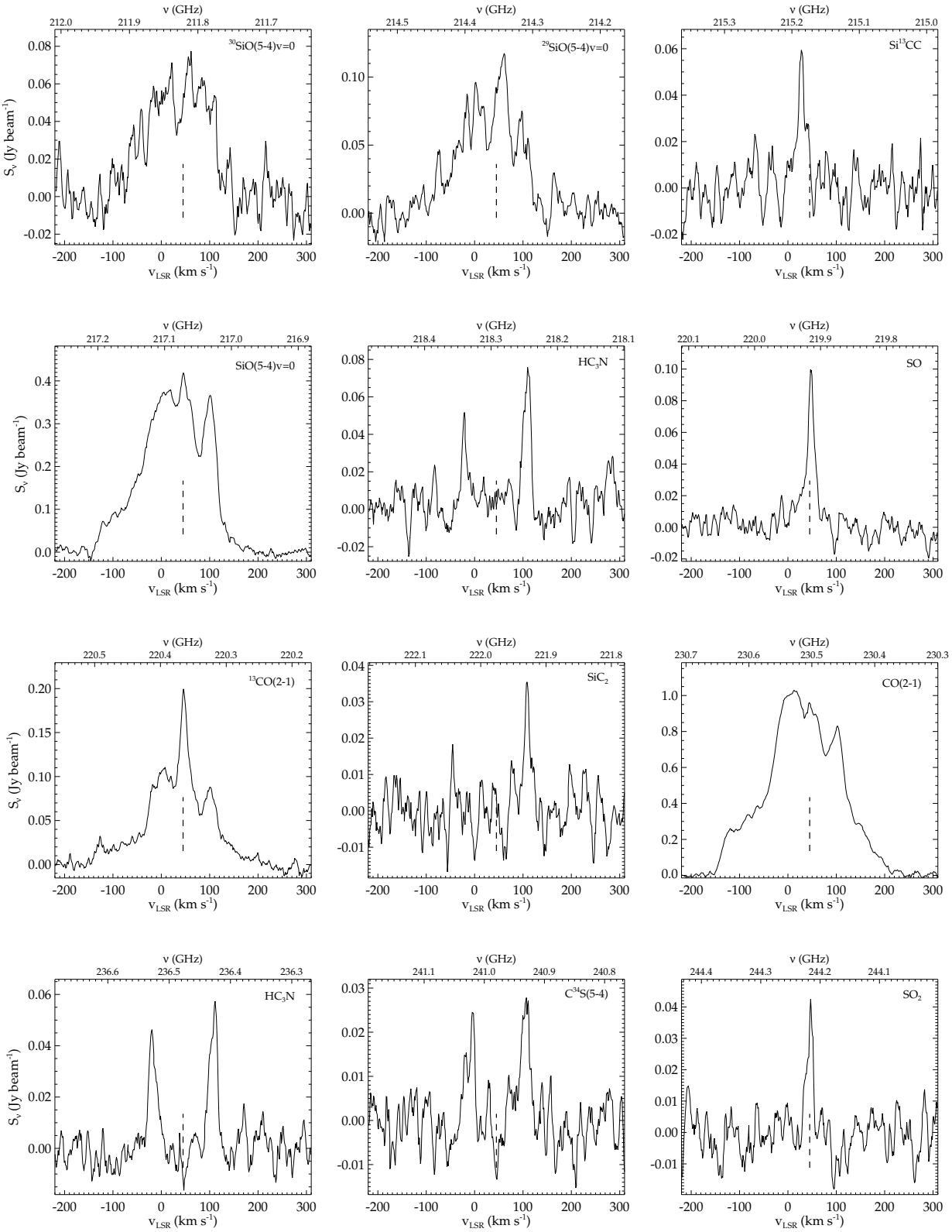

Figure 26. Spectral lines detected towards V510 Pup at 1.3 mm. The dashed vertical line indicates the stellar velocity.

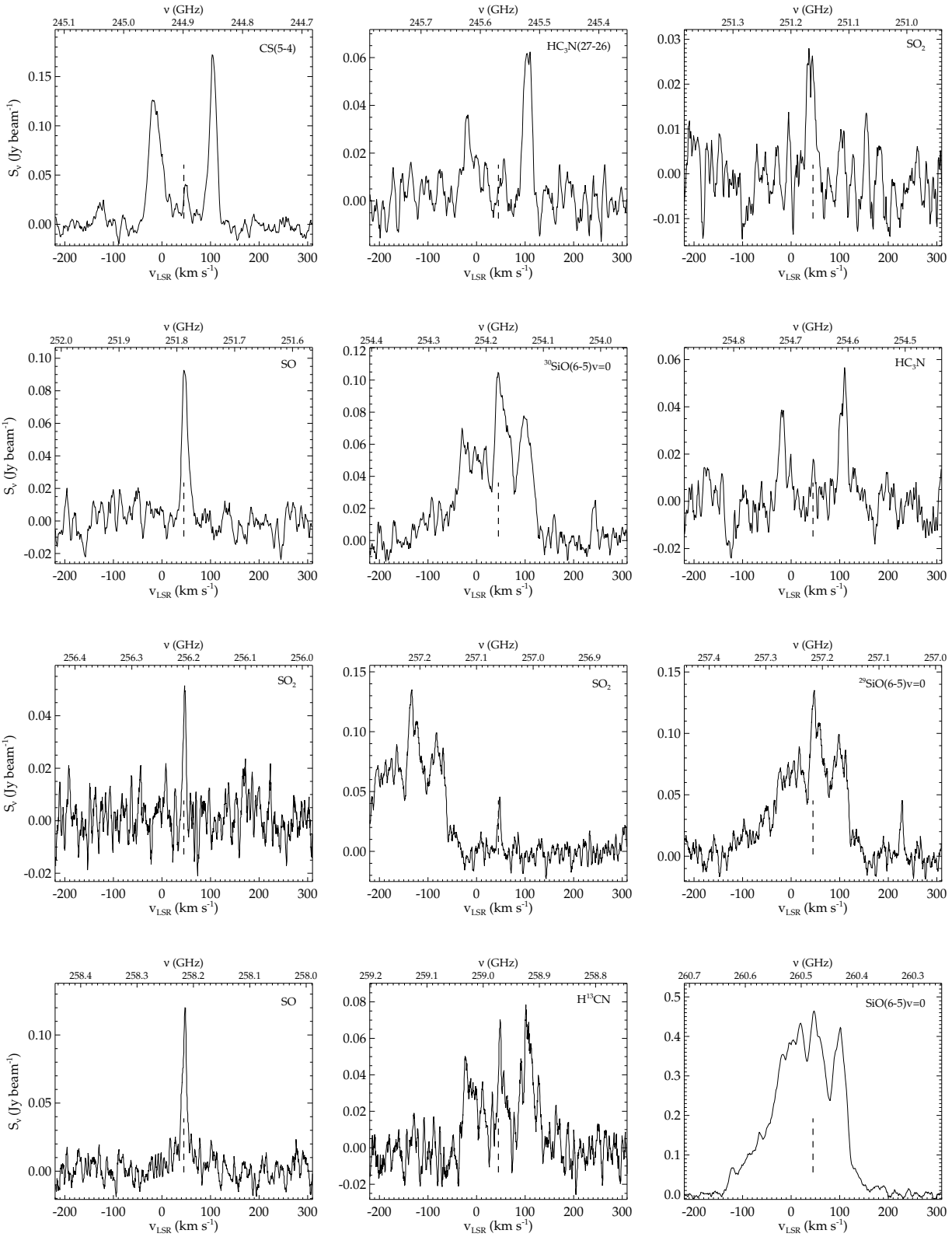

Figure 27. Spectral lines detected towards V510 Pup at 1.3 mm. The dashed vertical line indicates the stellar velocity.

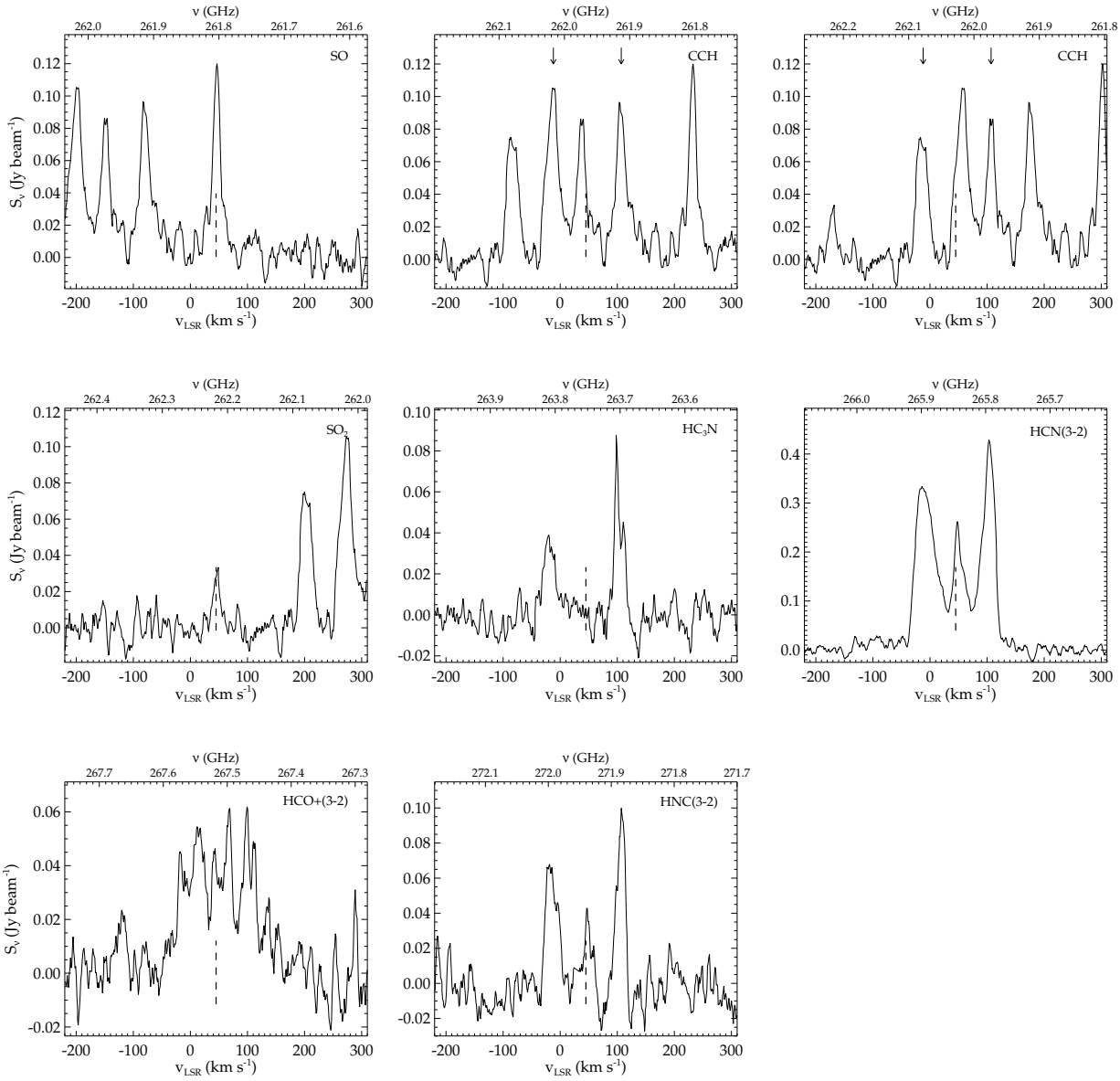

Figure 28. Spectral lines detected towards V510 Pup at 1.3 mm. The dashed vertical line indicates the stellar velocity.

Nothing here yet

Facilities: ALMA

Software:

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